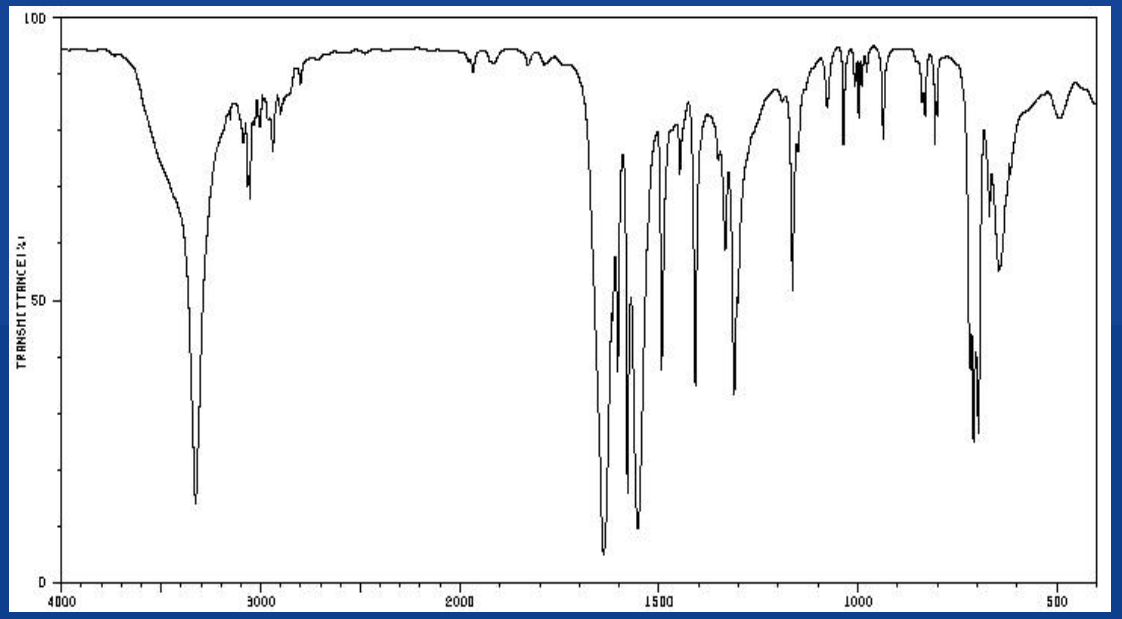
Analytical
Chemistry • Chapter 10

Infrared Absorption Spectroscopy (IR)



A Typical Infrared Absorption Spectrum

Ming Jiang

Contents

Four Sections and Key Points

Learning Path

- Understand the principle;
- Recognize the functional groups;
- Interpret the spectra.

01 Overview

What is an IR spectrum · Representation of spectra

02 Basic Principles of IR Absorption Spectroscopy

Molecular vibration · Modes of vibration · Characteristic and fingerprint regions

03 Typical IR Spectra of Organic Compounds

Aliphatic hydrocarbons · Aromatic hydrocarbons · Alcohols, phenols and ethers · Carbonyl and nitrogen compounds

04 IR Spectral Analysis

Preparation · Degree of unsaturation · Interpretation order and examples

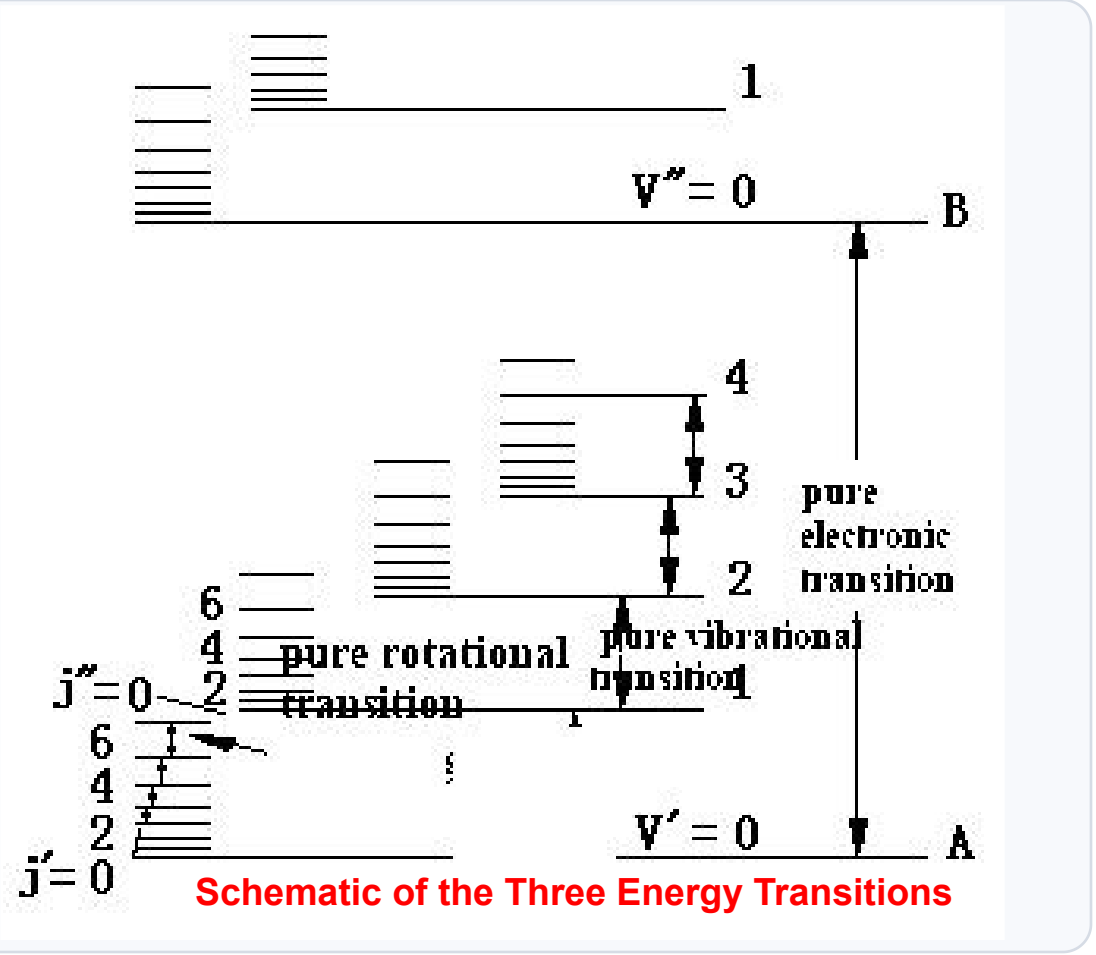
01

Section 1 · Overview

Introduction to Infrared Spectroscopy

- Energy transitions
- What is an IR spectrum
- Representation of spectra

Three Energy Transitions and Three Types of Spectra



Schematic of the Three Energy Transitions

Electronic transition → UV-Vis spectroscopy

ΔE ≈ 1—20 eV

Vibration-rotation transition → IR (this chapter)

ΔE ≈ 0.05—1 eV

Rotational transition → microwave spectroscopy

ΔE < 0.05 eV

A vibrational transition is always accompanied by a rotational transition, so the IR spectrum is also called a "vibration-rotation spectrum".

Section 1 • Overview

What Is an Infrared Spectrum

When a substance is irradiated with infrared light spanning a continuous range of wavelengths, the substance absorbs light in the infrared region; the spectrum produced by the vibrational-rotational energy transitions of its molecules is called an infrared spectrum (IR), also known as a vibration-rotation spectrum.

1

Irradiate

IR light of continuous wavelength selective absorption by the molecules

2

Transition

vibrational level + rotational level

$E_{vib} \gg E_{rot}$

3

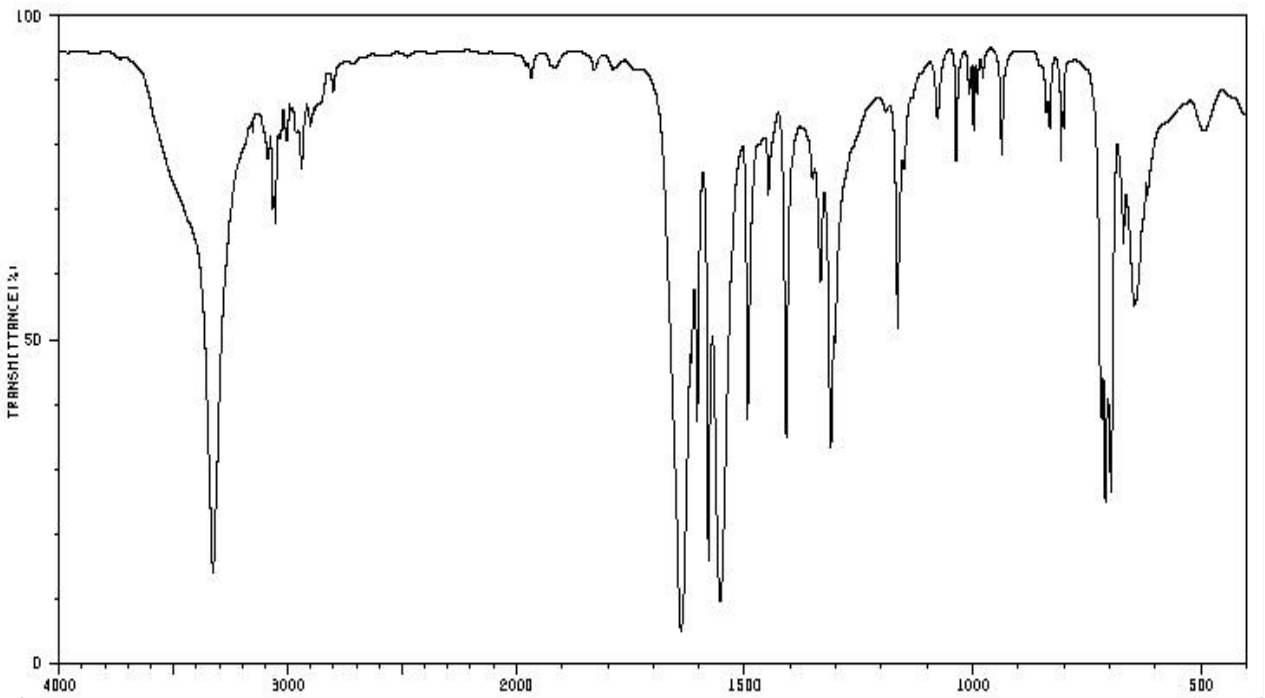
Record

absorption intensity vs wavenumber

obtaining the infrared spectrum

Section 1 • Overview

Representation of an Infrared Spectrum



Abscissa wavenumber σ (cm^{-1}), high on the left, low on the right

Ordinate transmittance T%

An IR spectrum uses wavenumber as the abscissa, which is proportional to energy; the ordinate may also be absorbance A.

Section 1 • Overview

Section Summary

1 Definition of an infrared spectrum

Irradiation with continuous-wavelength IR light → vibrational-rotational transition → IR (vibration-rotation) spectrum

2 Three energy transitions

Electronic level (UV-Vis) > vibrational level (IR) > rotational level (microwave)

3 Representation of spectra

Abscissa: wavenumber σ (4000→400); ordinate: transmittance T%; the troughs are the absorption peaks

Enter Section 2 with one question: why do molecules absorb exactly "certain" wavenumbers of infrared light?

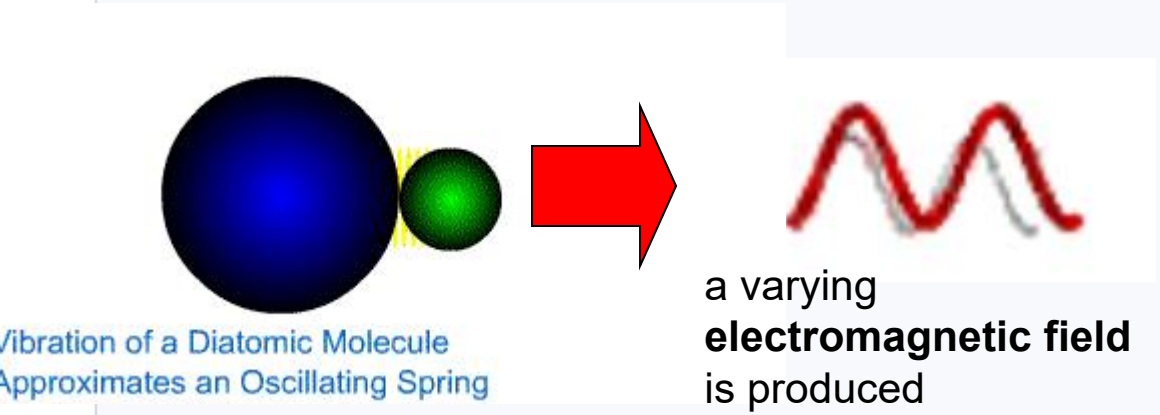
02

Section 2 • Basic Principles of IR Absorption Spectroscopy

• Molecular vibrational levels • Modes and degrees of freedom • Conditions for absorption

• Peak position and intensity • Characteristic and fingerprint regions • Characteristic and correlation peaks

Harmonic Oscillator Model: a Chemical Bond Is Like a Spring



a varying **electromagnetic field** is produced

The stiffness of the spring → the force constant k; the mass of the balls → the reduced mass μ.

① Vibration of a diatomic molecule

A vibrating chemical bond resembles two balls joined by a spring: the balls represent atoms and the spring the chemical bond.

② Vibration produces an alternating electromagnetic field

Vibration makes the dipole moment of the bond change periodically, producing an alternating electromagnetic field — the origin of its interaction with infrared light.

Section 2 • Basic Principles

Molecular Vibrational Levels Are Quantized

$$E_V = \left(V + \frac{1}{2}\right) h\nu$$

V — the vibrational quantum number

V = 0, 1, 2, 3 …
can only take discrete integers, so the levels are separated "step by step".

ν — the vibrational frequency of the chemical bond

determined by the properties of the bond itself (the force constant k and the reduced mass μ).

Energy remains even at V = 0

Zero-point energy $E_0 = \frac{1}{2}h\nu$: a bond is never at absolute rest, and vibration always persists.

Condition for the Vibrational-Rotational Transition: the Energies Must "Match"

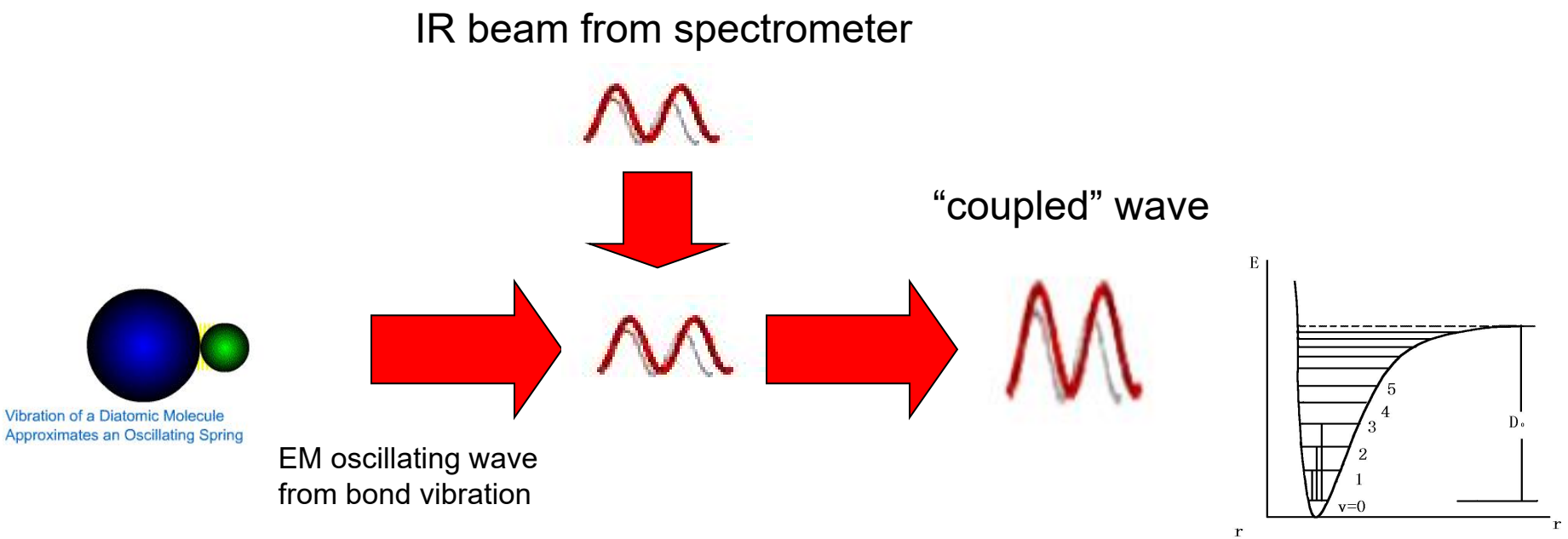
The difference between different vibrational levels is

$\Delta E_V = \Delta V\, h\, \nu$

Energy of light EL = h vL

That is, when ΔEV = EL or νL = ΔV·ν, a vibrational-rotational transition may occur.

$$EV = (V+1/2)h$$



Fundamental and Overtone Peaks

Fundamental peak

produced by V = 0 → 1, i.e. ΔV = ±1.

The absorption frequency equals the vibrational frequency ν of the bond itself; it is strong and is the main basis for interpreting a spectrum.

Example: ketone, νC=O ≈ 1715 cm^{-1}

Overtone peak

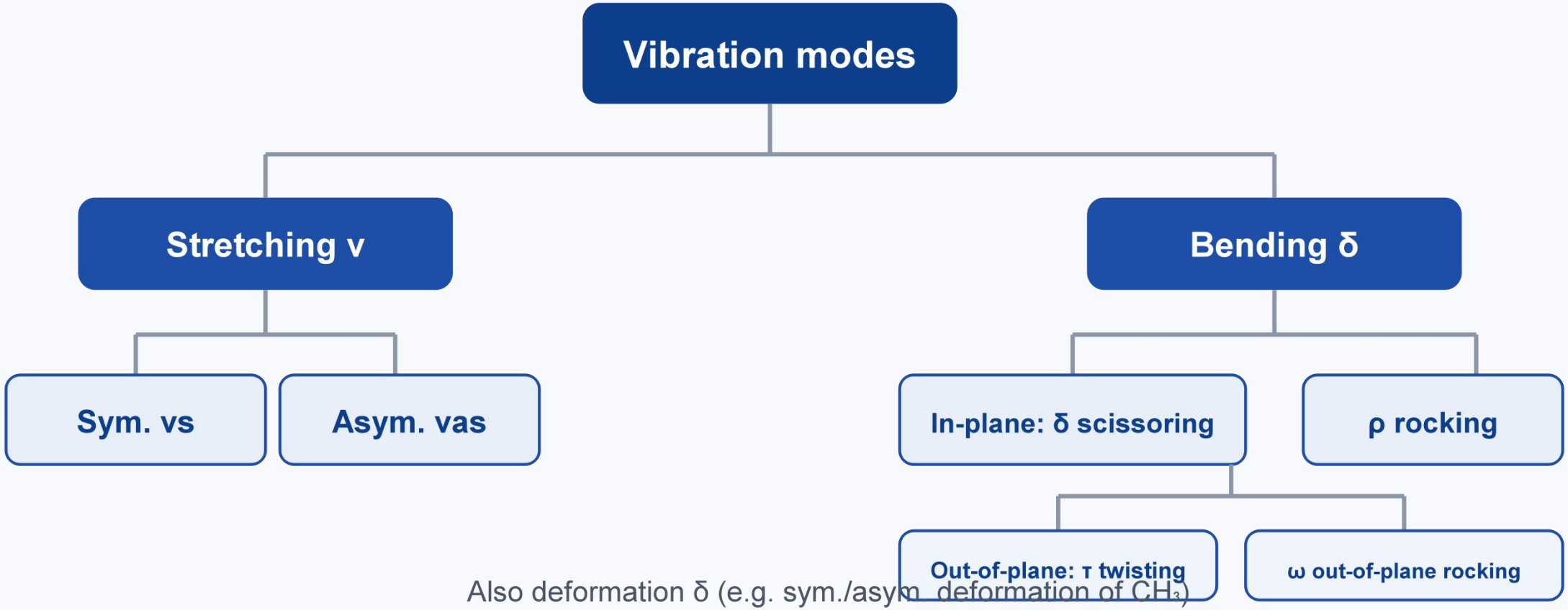
produced by V = 0 → 2, 3 ..., i.e. ΔV = ±2, ±3 ...

appears at 2 and 3 times the fundamental; it is weak and is also called an overtone peak.

Example: ketone, 1715 → overtone ≈ 3430 cm^{-1}

An overtone is weak, but when it is close to a fundamental it is "amplified" — this is "Fermi resonance".

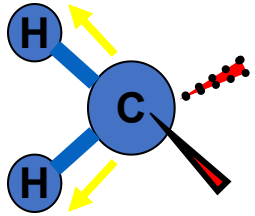
Overview of Vibrational Modes



There are two types of bond vibration:

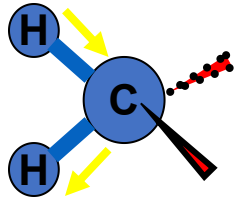
- ***Stretch*** **ν**





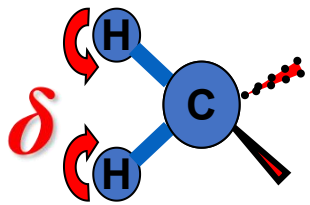
symmetric ν^{s}



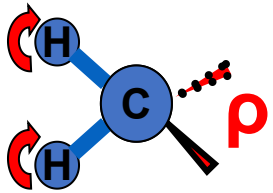


asymmetric ν^{as}

- ***Bend***

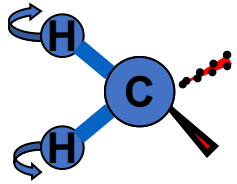


scissor

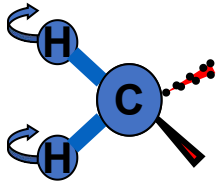


rock

in plane



twist



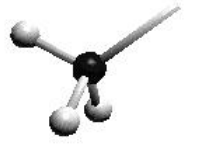
wag

out of plane



β

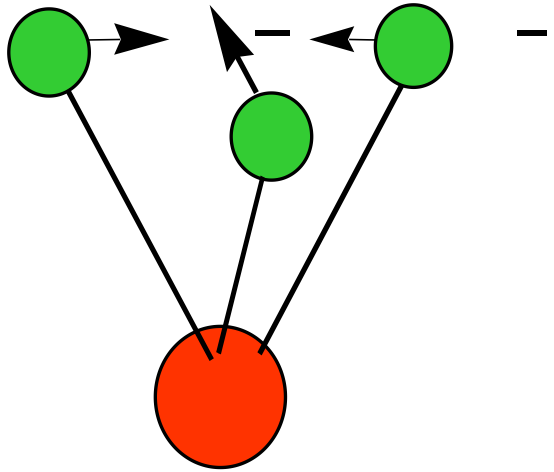


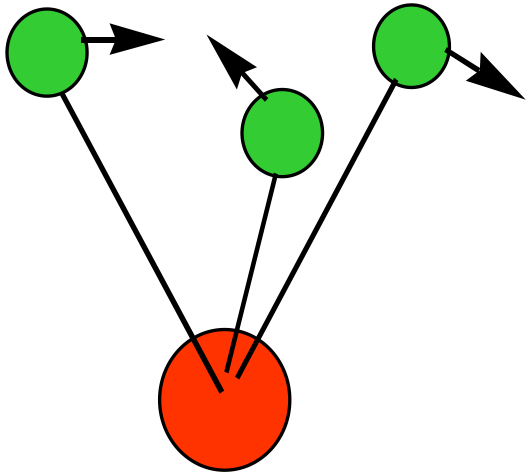


Deformation vibration

Deformation vibration

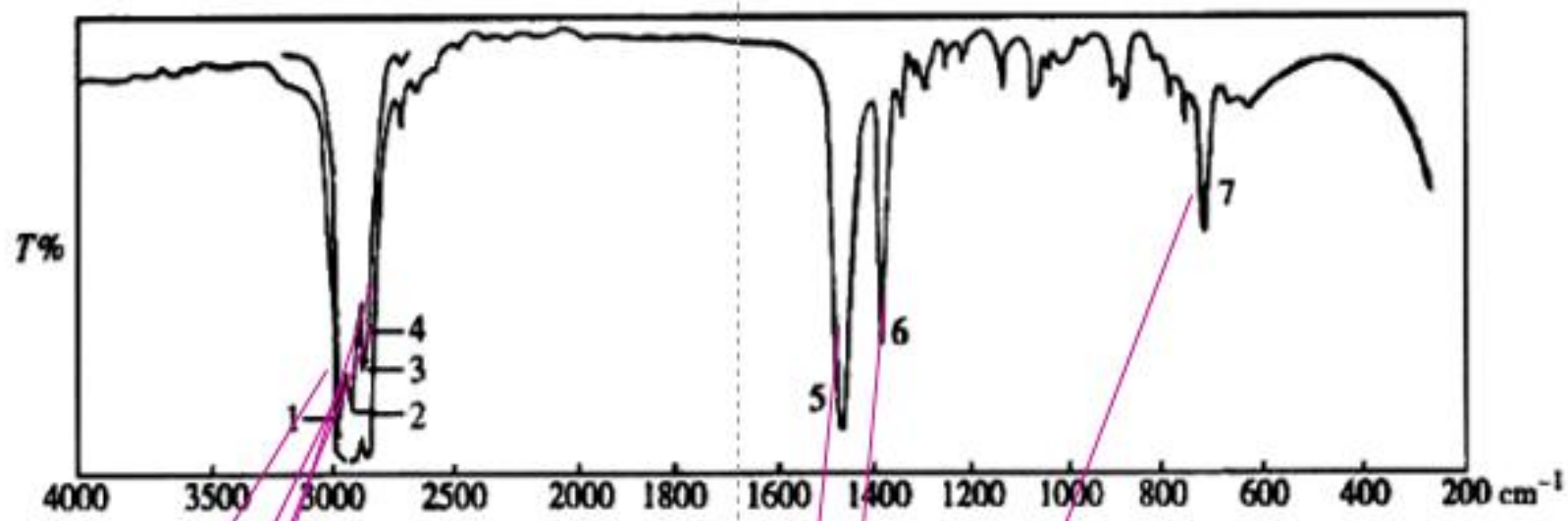
Methyl





symmetric δs(CH_3) 1380 cm^{-1}

asymmetric δas(CH_3) 1460 cm^{-1}



IR spectrum of n-hexane

$\nu^{as}_{CH_3} \sim 2960 cm^{-1}$

$\nu^{as}_{CH_2} \sim 2925 cm^{-1}$

$\nu^{s}_{CH_3} \sim 2870 cm^{-1}$

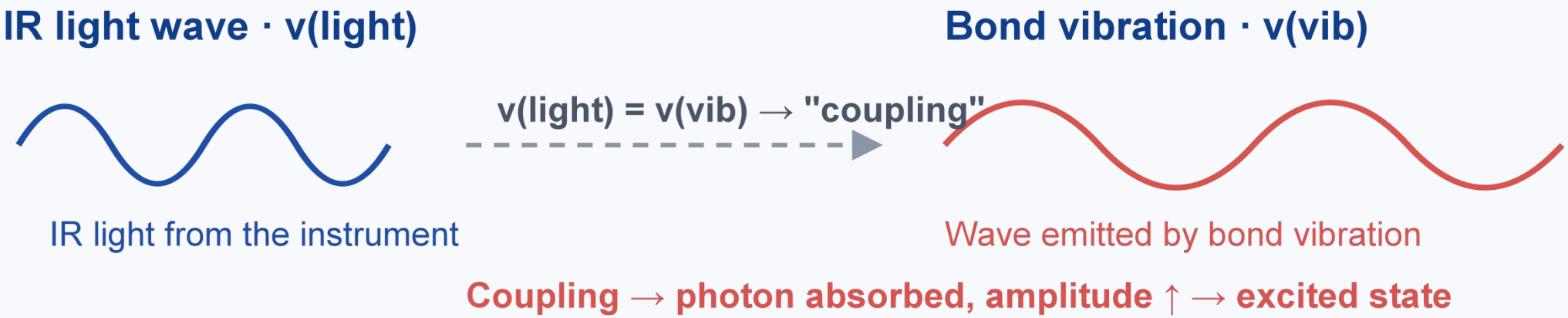
$\nu^{s}_{CH_2} \sim 2850 cm^{-1}$

$\delta^{as}_{CH_3} \sim 1450 cm^{-1}$

$\delta^{s}_{CH_3} \sim 1375 cm^{-1}$

$\omega_{CH_2} \sim 720 cm^{-1}$

Condition ① for IR Absorption: Frequency Matching



Condition 1 (necessary condition)

The frequency of the infrared light exactly matches the vibrational frequency of the bond, i.e. νlight = ΔV·ν.

Analogy

Pushing a swing: you must push at its natural rhythm to make it swing higher; if the frequency is wrong, no energy is transferred.

Condition ②: the vibration must change the dipole moment (Δμ ≠ 0)

IR-active: H_2O

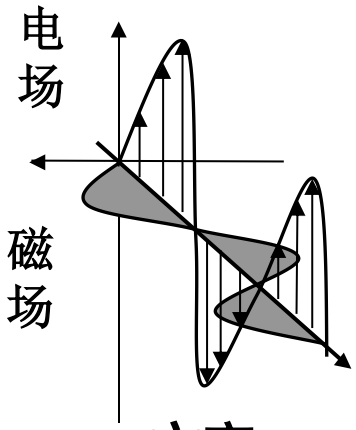
The O—H bond is polar; its vibration changes the distance between the positive and negative charge centers, Δμ ≠ 0.

Symmetric stretch, asymmetric stretch and bending all change the dipole moment → all three vibrations give peaks.

IR-inactive: the symmetric stretch of CO_2

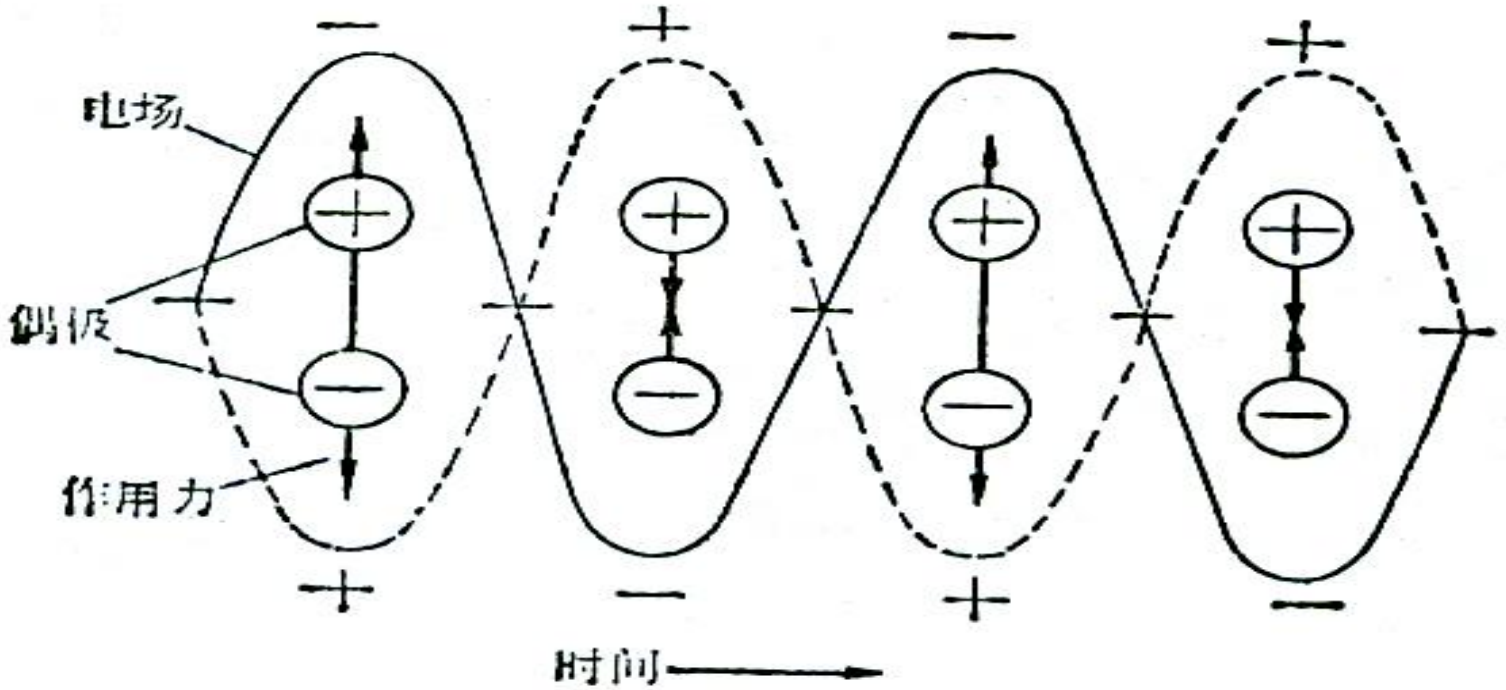
The two ends of O=C=O are symmetric; during the symmetric stretch the dipole changes of the two C=O bonds cancel out, Δμ = 0.

This vibration produces no infrared absorption (it can be observed in the Raman spectrum).



交变磁场

ν



What Determines the Intensity of an Absorption Peak?

Peak intensity ∝ transition probability ∝ the magnitude of the dipole-moment change during vibration (Δμ)

Strongly polar bonds give strong peaks

Strongly polar bonds such as C=O, O—H and C≡N have a large Δμ → strong peaks; a symmetric C—C → weak or no peak.

Fundamental > overtone

The fundamental transition is far more probable than the overtone, so overtone peaks are weak (vw / w).

Convention of Intensity Symbols

vs very strong · s strong · m medium · w weak · vw very weak · b broad · sh sharp — used throughout this book.

Fundamental Vibrational Frequency: Hooke's Law

Hooke's Law

$$\nu(\text{frequency}) = (1/2\pi)\cdot\sqrt{(k/\mu)}$$
$$\text{or} \quad \nu(\text{wavenumber}) = (1/2\pi c)\cdot\sqrt{(k/\mu)}$$

k — the force constant of the chemical bond

Unit: $N\cdot cm^{-1}$. The stronger the bond (larger bond energy, shorter bond), the larger k and the higher the frequency.

μ — the reduced mass

Unit: g. $\mu = m_1m_2/(m_1+m_2)$; the lighter the atoms, the higher the frequency.

One-Line Memory Aid

"Stronger bond, higher frequency; lighter bond, higher frequency" — as k increases or μ decreases, the peak moves to higher wavenumber.

Discussion ①: The Larger the Bond Energy, the Higher the Frequency

Type of bond	C≡C	C=C	C—C
Force constant k	12~18	8~12	4~6
σ / cm^{-1}	2100~2260	1620~1680	700~1200

k is in units of $N \cdot cm^{-1}$ (order of magnitude $\times 10^{-5}$)

The stiffer the "spring" of the bond, the faster it vibrates and the further to the left (higher wavenumber) the absorption peak lies.

Conclusion

Bond energy ↑ → k ↑ → ν or σ ↑ triple bond > double bond > single bond.

Discussion ②: The Lighter the Atom, the Higher the Frequency

Chemical bond	C—H	C—N	C—O
σ / cm^{-1}	**2853~2960**	**1180~1360**	**1080~1300**
Bonded atoms	**H is the lightest**	N	O

The three bonds have similar k; the difference comes mainly from the magnitude of μ

Conclusion

As the mass of the bonded atoms m_1 or m_2 decreases → μ decreases → ν or σ increases.

Bonds containing H always lie in the high-wavenumber region.

This rule is the starting point for judging "who is absorbing" above 3000 cm^{-1}.

Question: Can Infrared Spectroscopy Distinguish Isotopes?

Reasoning

D is twice as heavy as H, so μ increases markedly while k stays essentially unchanged. Since $\nu \propto \sqrt{(k/\mu)}$, the wavenumber drops to about $1/\sqrt{2}$.

Conclusion

Yes, it can. Isotopic substitution causes a clear red shift of the characteristic peak and is a powerful tool for studying isotope effects.

Discussion ③: For the Same Group, ν > β > γ

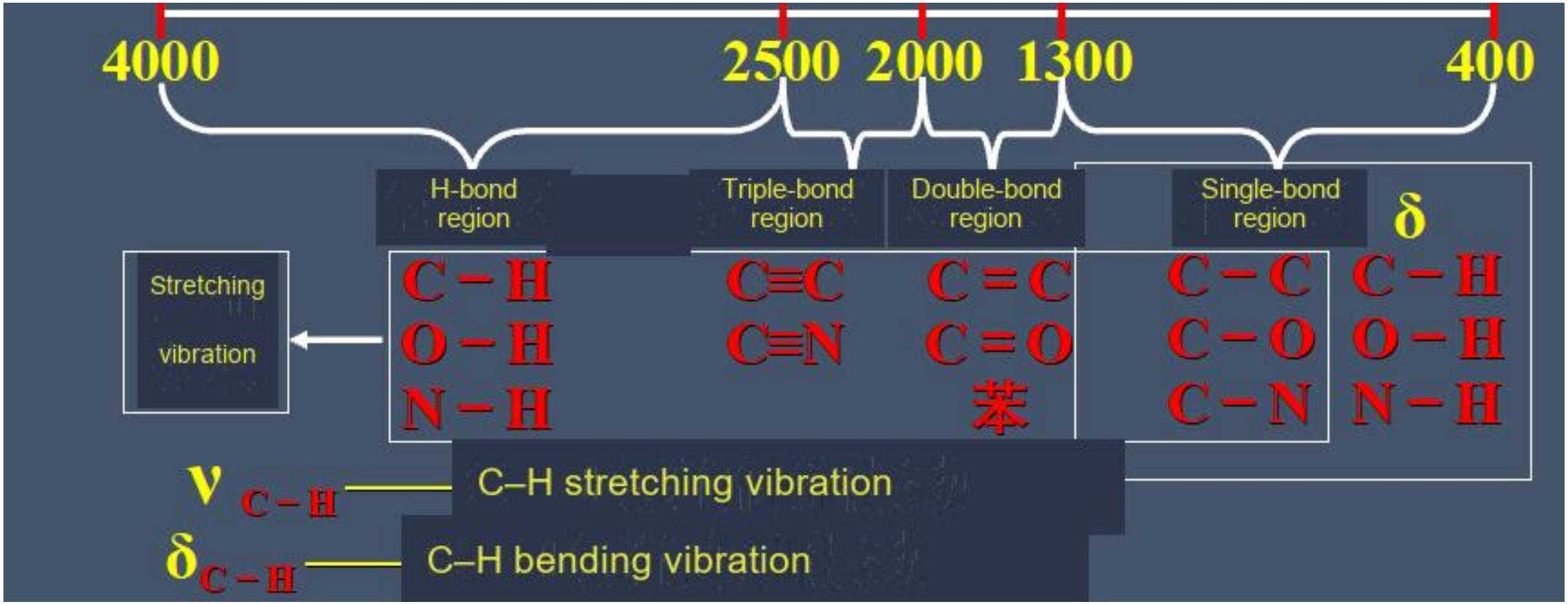
When the reduced mass is the same

ν (stretching) > β (in-plane bending) > γ (out-of-plane bending)

e.g. νC—H > βC—H > γC—H



Reason: Hooke's formula is based on stretching; "bending" a bond takes less effort — the force constant is smaller, so the frequency is lower.


4000
2500
2000
1300
400
H-bond region
Triple-bond region
Double-bond region
Single-bond region
δ
Stretching vibration
C – H
O – H
N – H
C≡C
C≡N
C = C
C = O
苯
C – C
C – O
C – N
C – H
O – H
N – H
ν C – H
C–H stretching vibration
δ C – H
C–H bending vibration

Factors Affecting Peak Position: an Overview

Inductive effect

electron-withdrawing → ν↑

Conjugation effect

electron delocalization → ν↓

Steric effect

conjugation hindered → ν↑

Ring-strain effect

smaller ring → ν↑

Hydrogen bonding

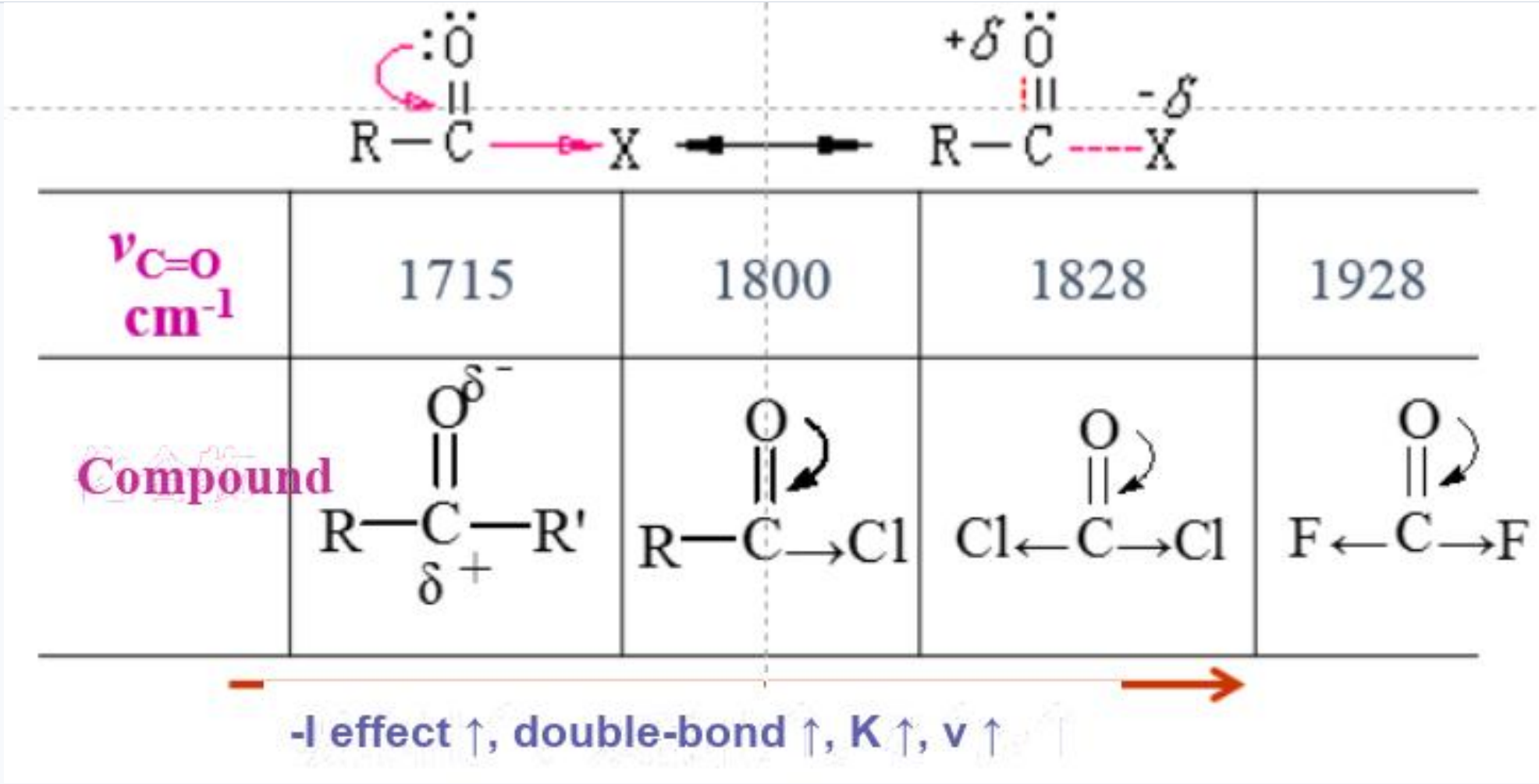
association → ν↓
and broadens

Fermi resonance / vibrational coupling

peak splitting and shift

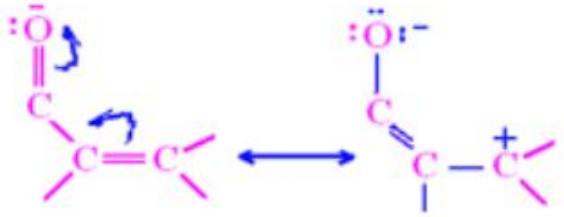
The fundamental frequency is determined by k and μ; but these "environmental factors" can shift the peak of the same functional group by tens to hundreds of cm^{-1} — they must be considered when interpreting a spectrum.

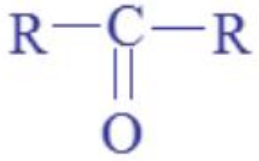
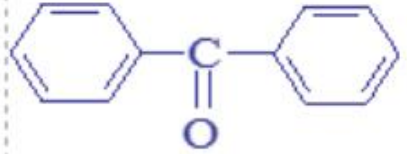
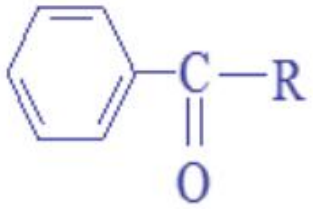
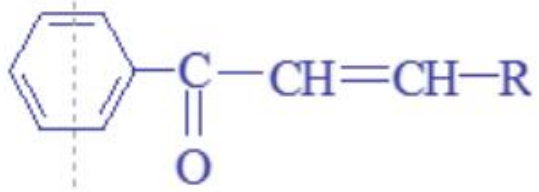
(1) Inductive Effect: Electron Withdrawal Raises ν



(2) Conjugation Effect: Electron Delocalization Lowers ν

Delocalization of π electrons over the conjugated system **(since K ↓)**, absorption shifts to **lower** wavenumber.

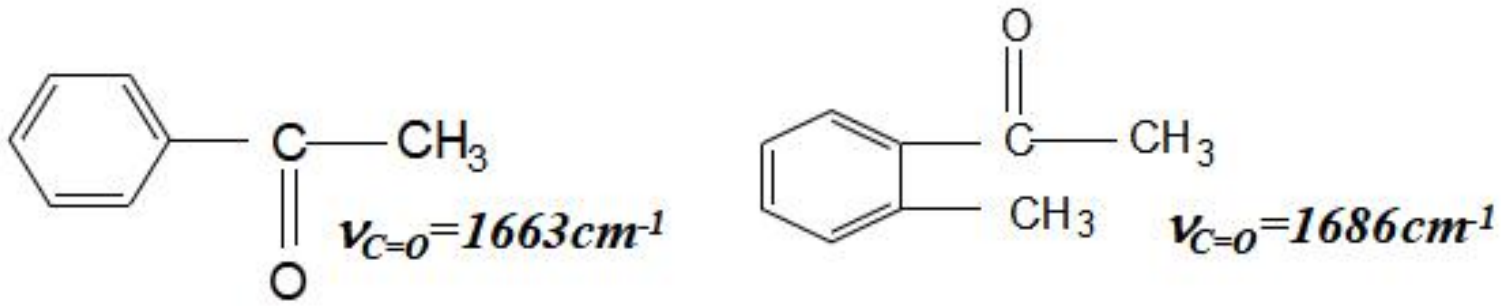


Compound	$\nu_{C=O}$ cm^{-1}	Compound	$\nu_{C=O}$ cm^{-1}
	1710～1725		1667～1661
	1695～1680		1667～1653

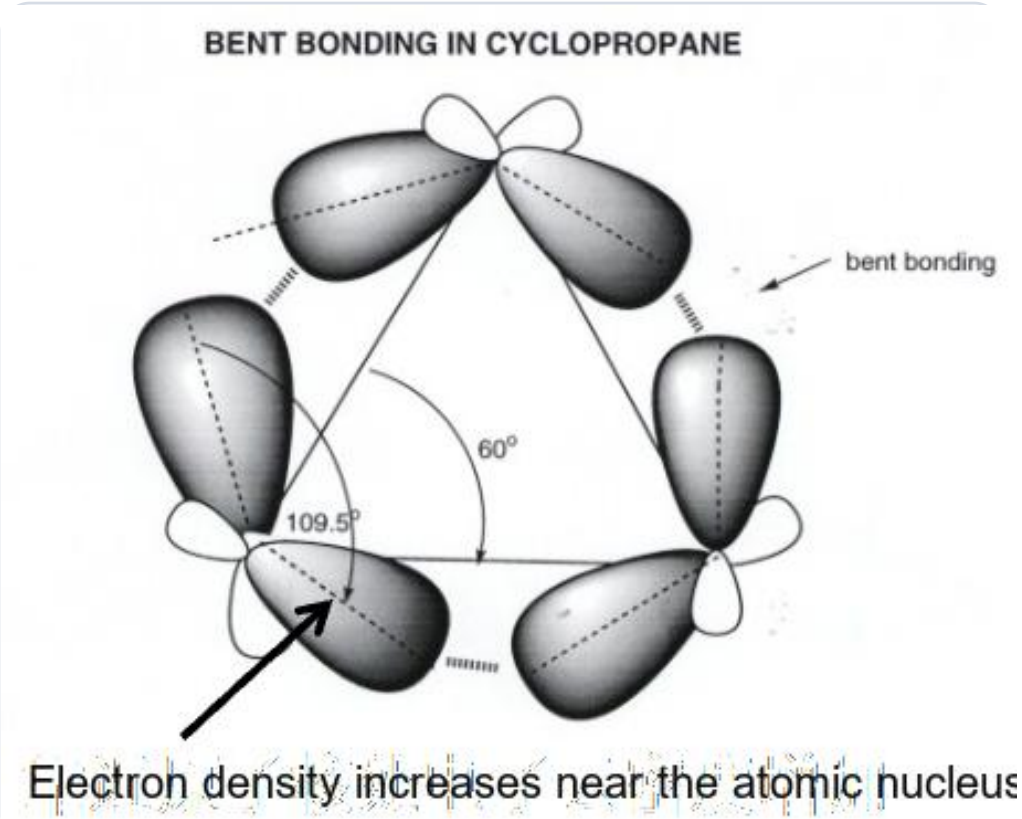
Conjugation ↓, double-bond character ↓, K ↓, ν ↓

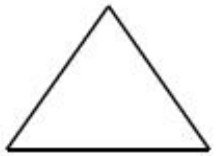
(3) Steric Effect: Conjugation Is "Broken Apart"

Because of steric hindrance, the molecular plane is no longer coplanar with the double bond, so the conjugation effect weakens — with reduced conjugation the double-bond character of the carbonyl is restored and the infrared peak shifts to higher wavenumber.



(4) Ring-Strain Effect





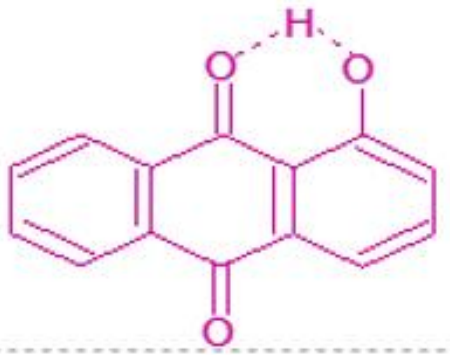
3060-3030 cm^{-1}



2900-2800 cm^{-1}

(5) Hydrogen Bonding: Association Red-Shifts and Broadens the Peak

Usually shifts the stretching frequency notably **lower**; the band becomes **broader**

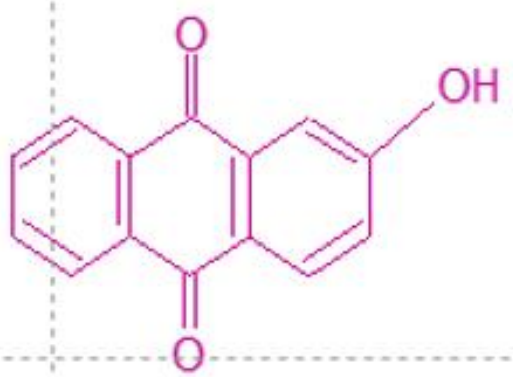


(intramolecular H-bond formed)

$\nu_{C=O}$(associated) 1622 cm^{-1}

(free) 1675 cm^{-1}

ν_{OH}(associated) 2843 cm^{-1}



(no intramolecular H-bond)

$\nu_{C=O}$(free) 1676 cm^{-1}

1673 cm^{-1}

ν_{OH}(free)

(3615~3615 cm^{-1})

▶ Intramolecular H-bond: large effect on peak position, independent of concentration

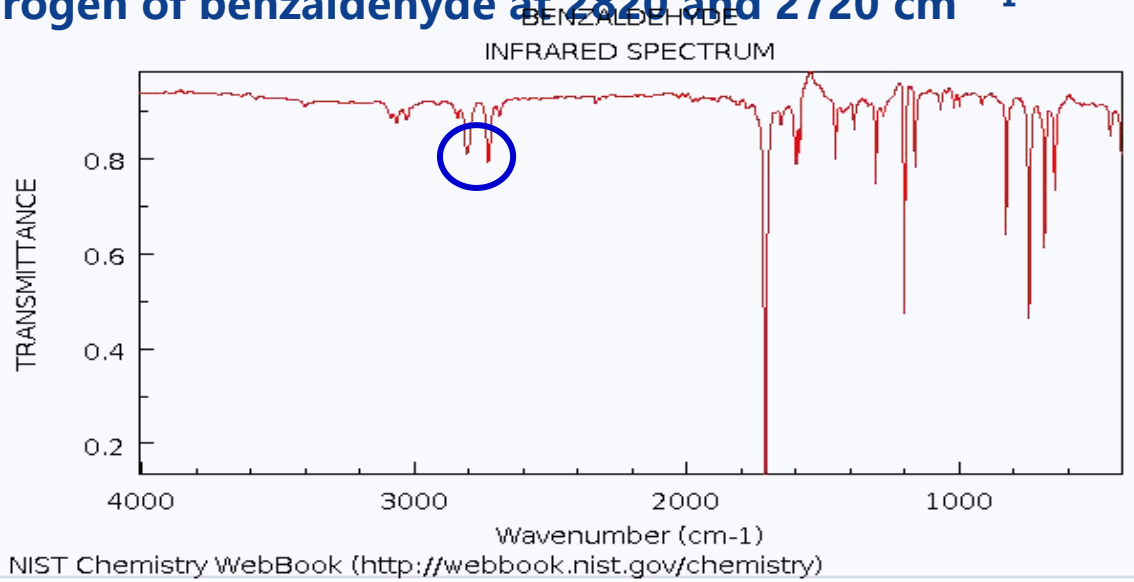
(6) Fermi Resonance

When the overtone of one vibration is close to the fundamental of another, coupling can occur — the overtone gains intensity or the peak splits; both the position and intensity change.

Classic Example: the "doublet" of the aldehyde hydrogen of benzaldehyde at 2820 and 2720 cm^{-1}

aldehyde νC—H ≈ 2850 cm^{-1}

\+ the overtone of δC—H at 1390 ≈ 2780 cm^{-1}



The two are close → Fermi resonance → splitting into two strong peaks at 2820 and 2720 (not two kinds of hydrogen!)

(7) Vibrational Coupling: One Becomes Two

When two groups with the same or similar vibrational frequencies are adjacent and share a common atom, the frequencies change: one moves to higher frequency and the other to lower — the band splits.

Example 1: Acid anhydride

Two C=O groups share one O: coupling splits into **a doublet at νas 1850 and νs 1780 cm^{-1}.**

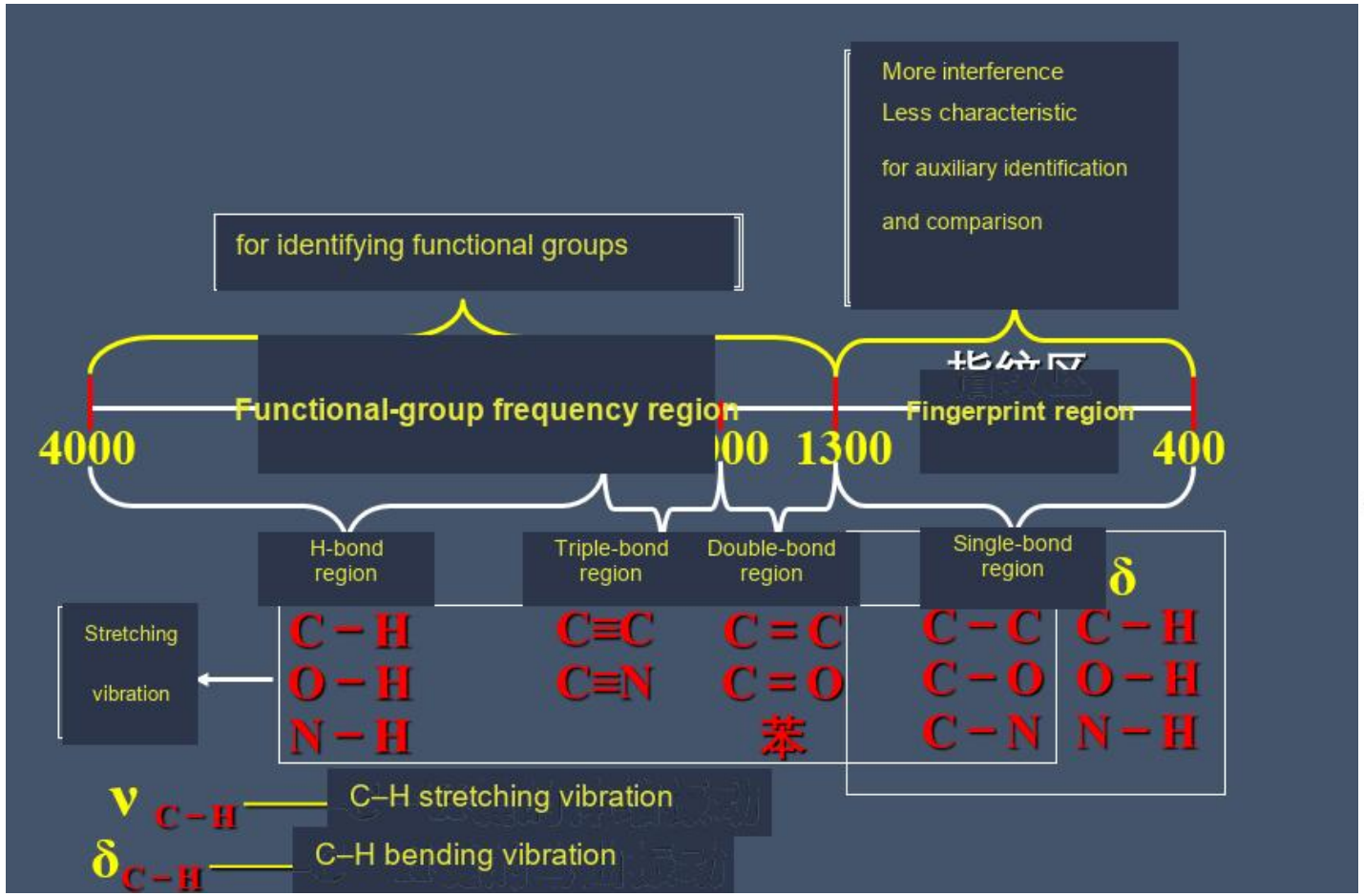
Example 2: Isopropyl / tert-butyl

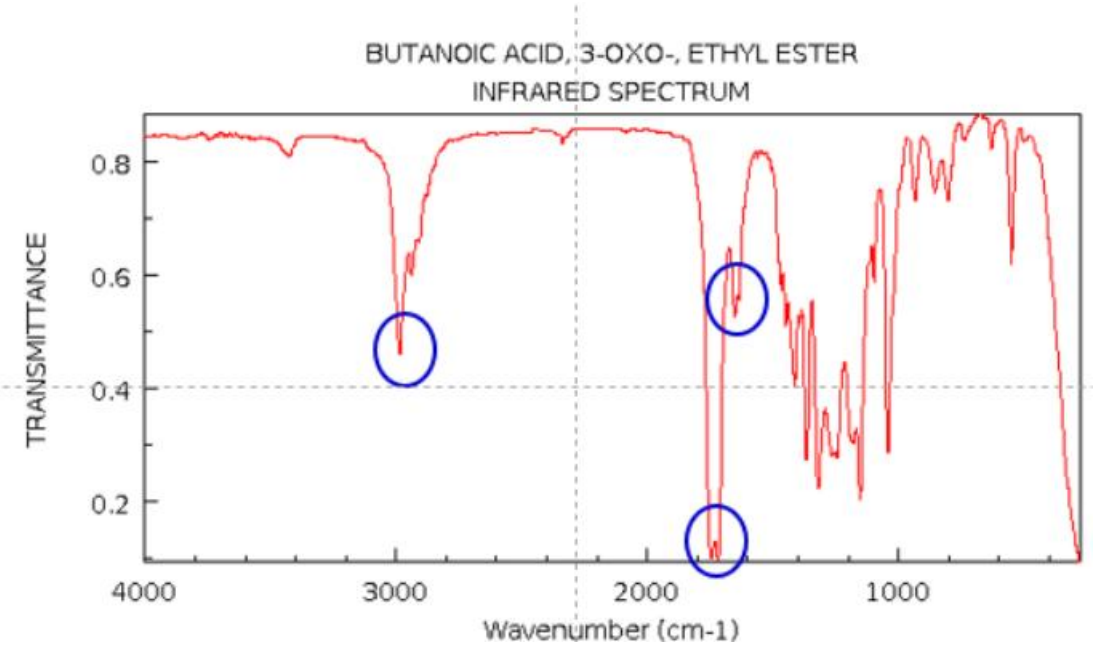
Coupling of the deformation vibrations of adjacent CH_3 groups: the 1380 peak splits into a doublet at 1385 and 1370 (isopropyl).

Difference from Fermi Resonance

Coupling occurs between two fundamentals; Fermi resonance is coupling between a fundamental and an overtone.

Characteristic and Fingerprint Regions: Reading the Spectrum in Two Halves





Characteristic Region (4000~1300 cm^{-1}), the High-Frequency Region

absorption peaks are sparse, fairly strong and easy to recognize

— the "house numbers" are clear

Role in interpretation:

judging which functional groups the compound contains

This region mainly contains the stretching vibrations of hydrogen-bearing groups (O—H, N—H, C—H) and of multiple bonds (C≡C, C≡N, C=O, C=C) — qualitative identification of functional groups relies on it.

Fingerprint Region (1300~400 cm^{-1}), the Low-Frequency Region

Features

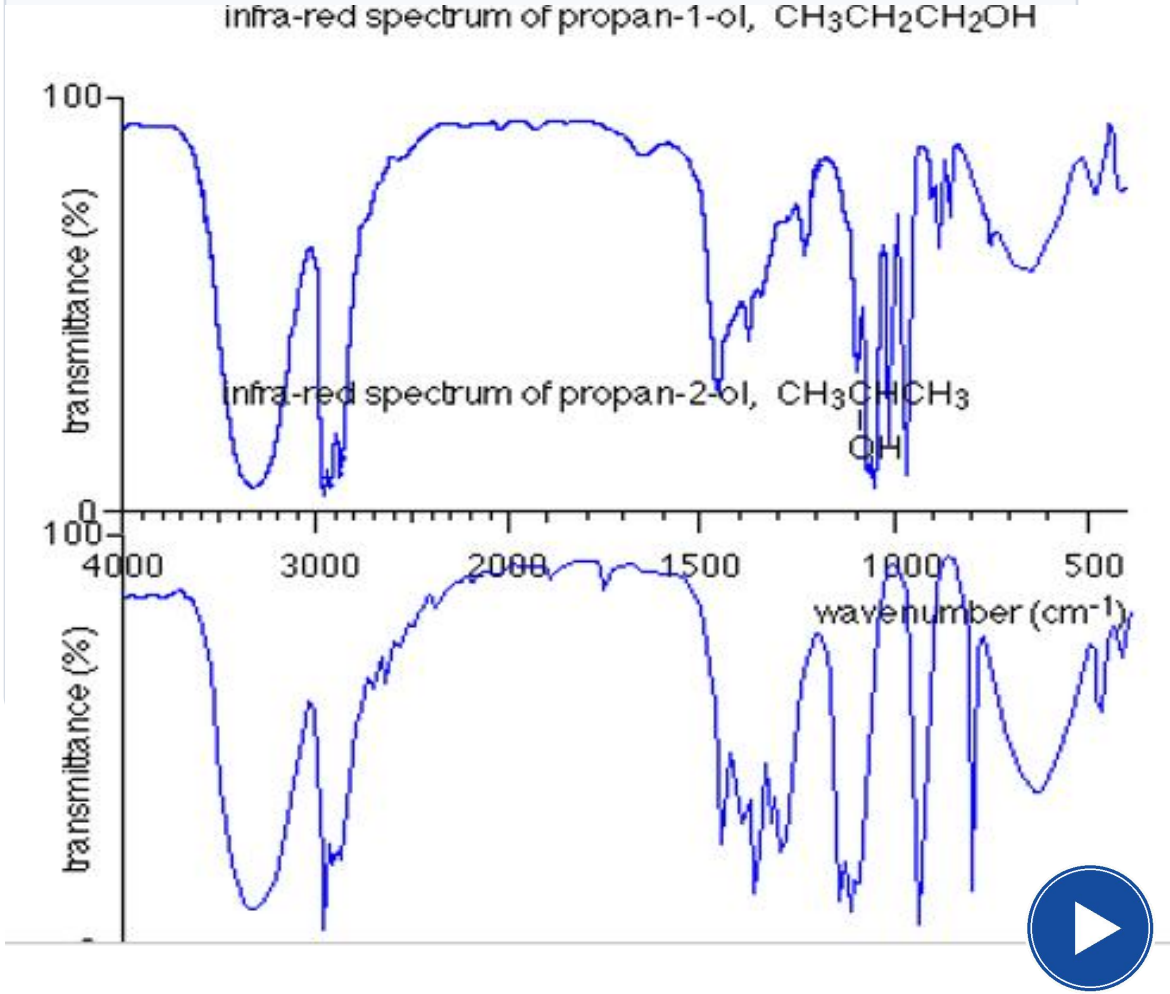
The peaks are dense and hard to recognize, like a human fingerprint — a slight difference in molecular structure produces a markedly different spectrum here.

Role in interpretation

① as supporting evidence for which functional groups the compound contains;

② to determine the finer structure of the compound (substitution pattern, differences in the skeleton, etc.).

Typical application: fingerprint comparison



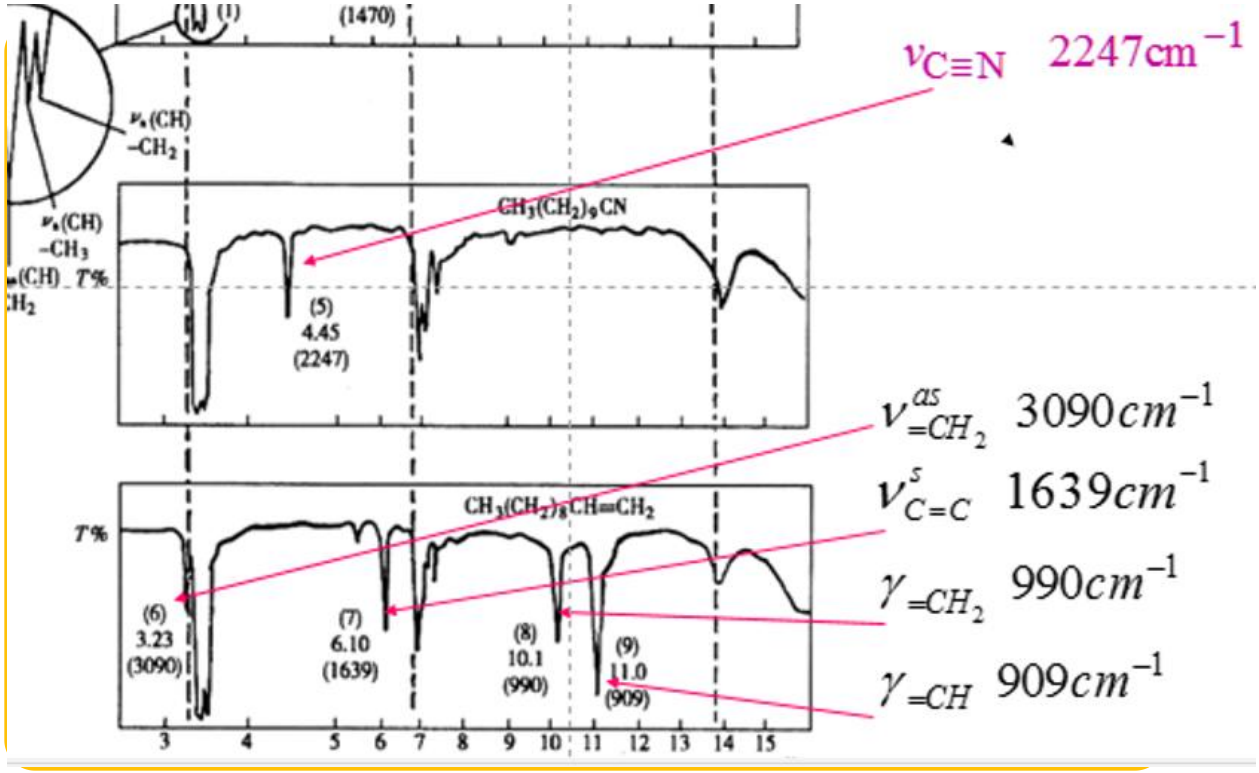
Characteristic Peaks and Correlation Peaks

Characteristic peak

Any absorption peak that can be used to identify the presence of a functional group.

Correlation peaks

A set of mutually dependent absorption peaks produced by the presence of a single functional group.



Note: the presence of a functional group can be confirmed only by using a set of correlation peaks.

03

Section 3 • Typical IR Spectra of Organic Compounds

• Aliphatic hydrocarbons • Aromatic hydrocarbons • Alcohols, phenols and ethers • Carbonyl compounds • Nitrogen compounds

Learning Method: Three Steps

1

Identify the characteristic structure

What does this functional group look like? What bonds does it have?

2

Analyze the relevant vibrations

What stretching and bending modes does each bond have?

3

Memorize the characteristic frequencies

turn the key wavenumbers into a "conditioned reflex"

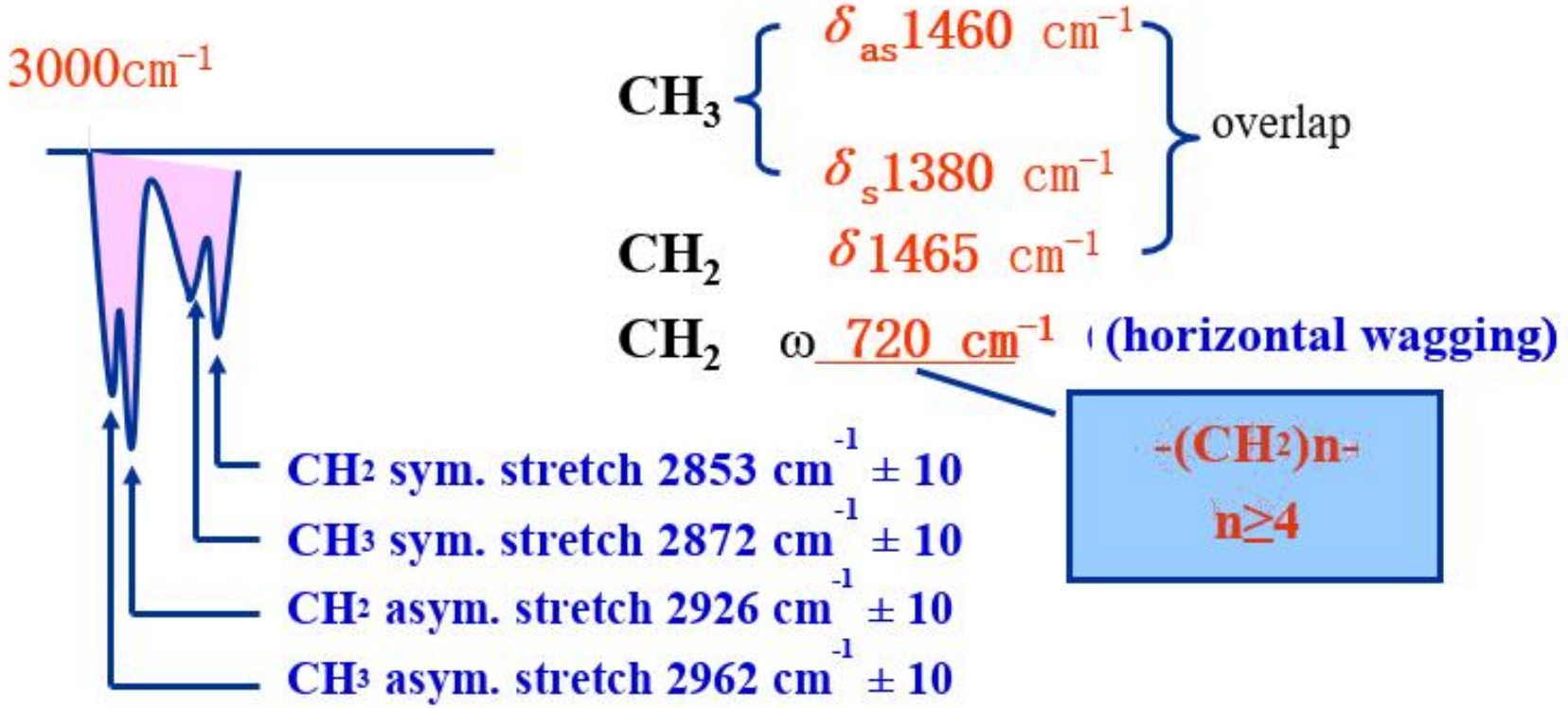
This section proceeds class by class in the order "aliphatic → aromatic → alcohols, phenols and ethers → carbonyl compounds → nitrogen compounds", each illustrated with real spectra.

Alkanes

1. C–H stretching vibration

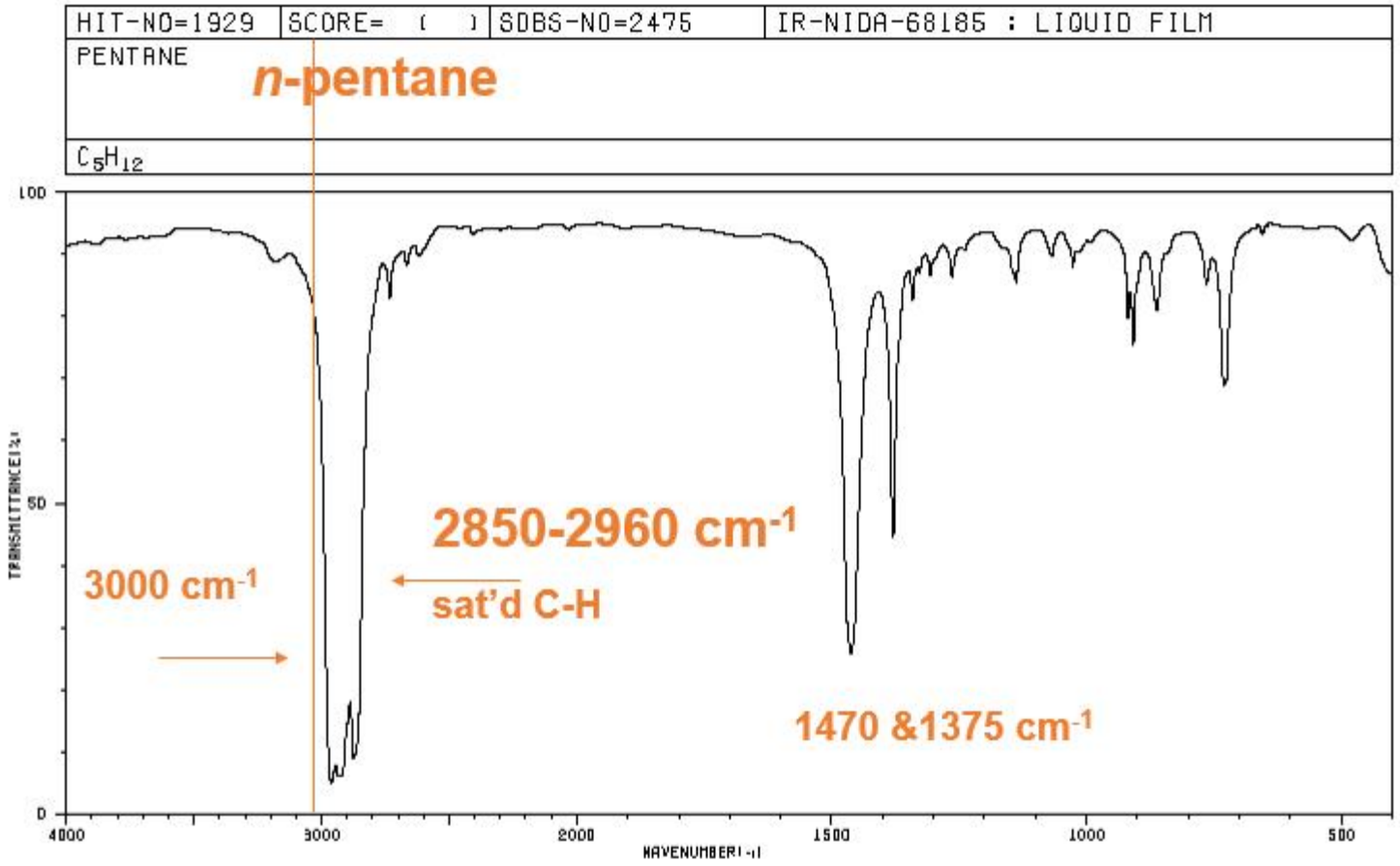
2. C–H bending vibration

3. C–C skeletal vibration ν_{C-C} 1250–1140 cm^{-1} (weak→medium) (poor specificity)



Alkanes

Example 1

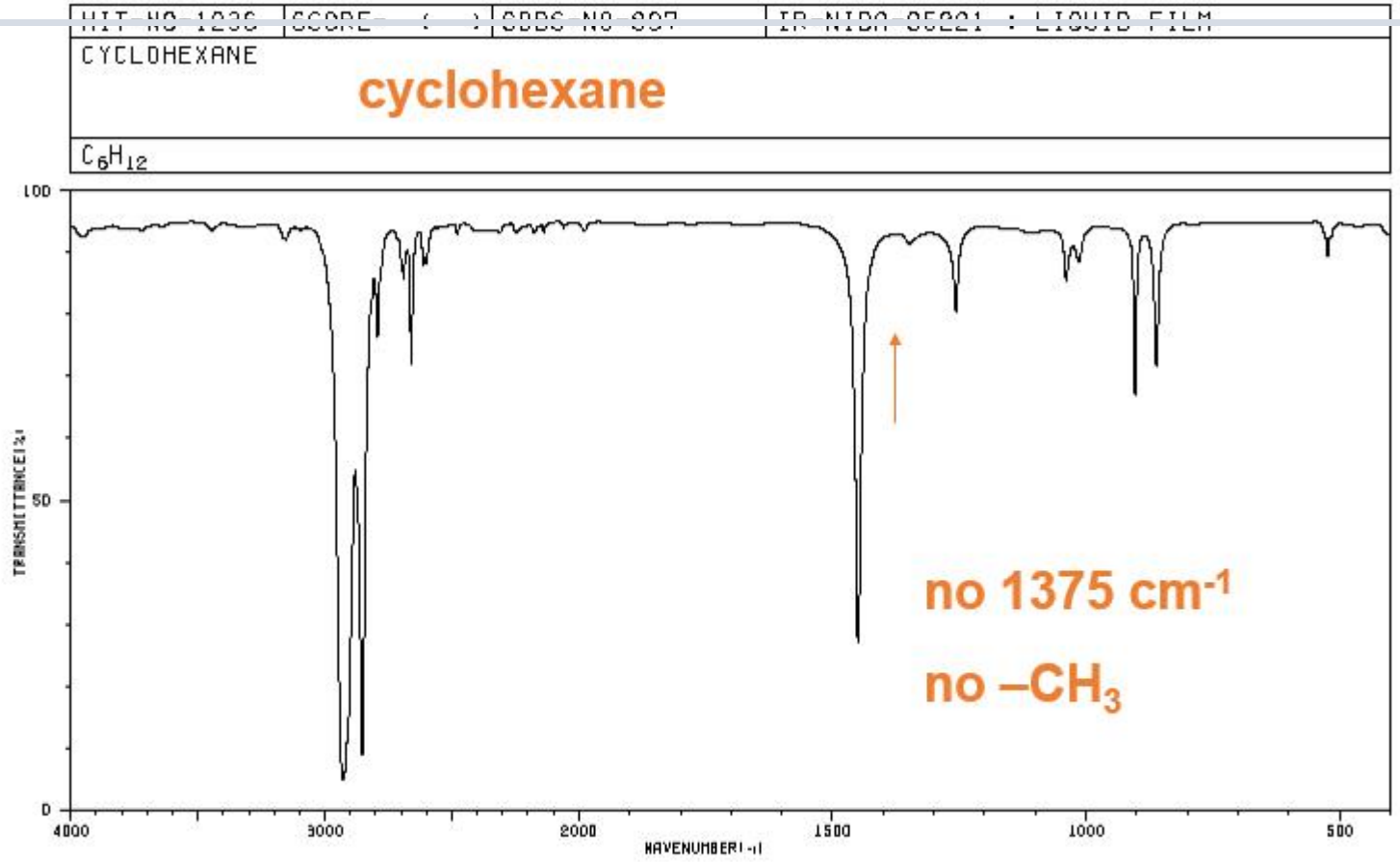


3182	86	2619	86	1138	81	732	66
3172	86	1462	24	1068	86		
2961	4	1380	45	1027	84		
2928	6	1342	79	919	77		
2875	8	1307	84	908	72		
2733	79	1264	84	862	79		
2666	84	1144	84	765	81		

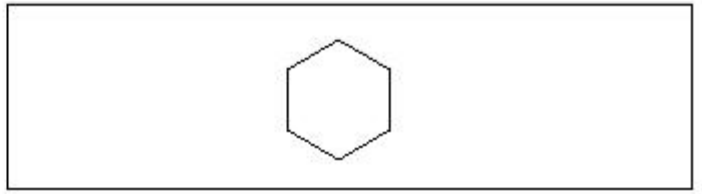
$CH_3CH_2CH_2CH_2CH_3$

Alkanes

Example 2



2928	4	1460	26
2853	8	1257	77
2794	72	1039	81
2690	81	1016	84
2661	70	904	64
2613	84	862	68
2599	84	624	86



Alkanes

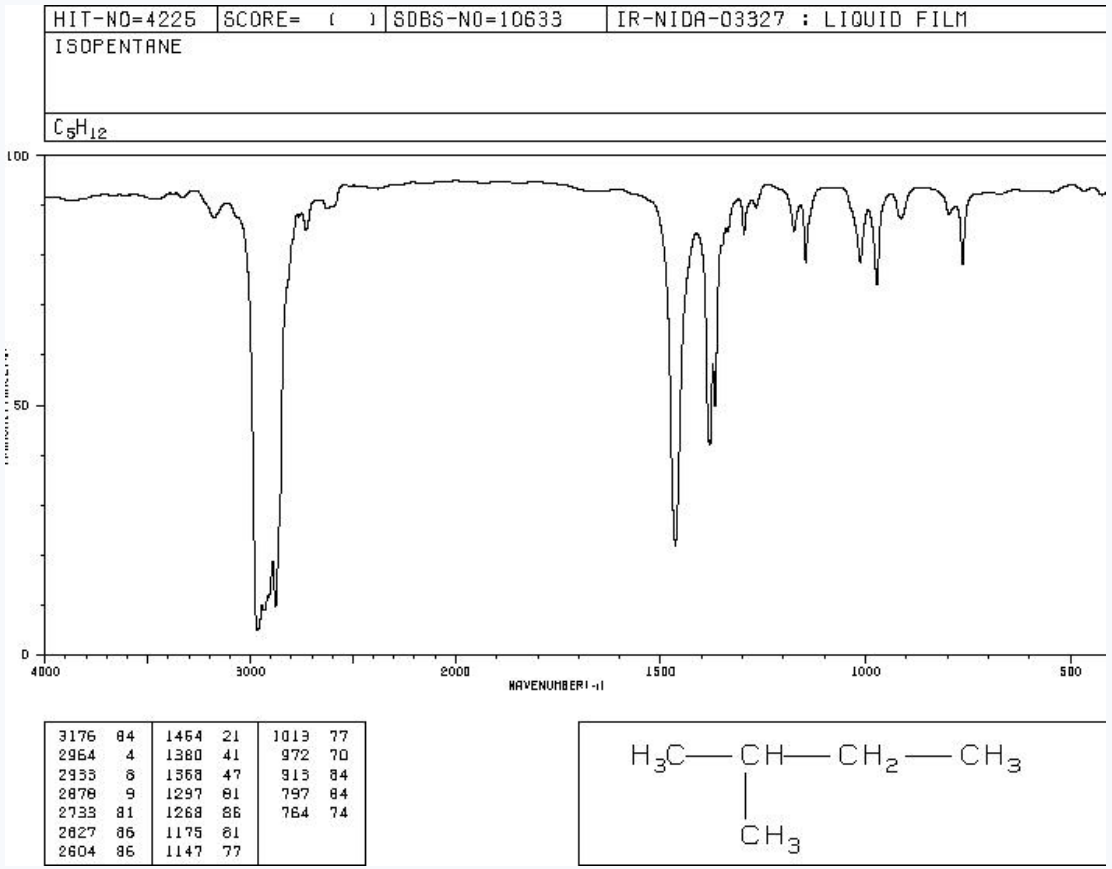
Using 1380 cm^{-1} to Judge the Methyl Environment

$-CH_2-$	+	1430-1470
$-CH_3$	+	" and 1375
$-CH(CH_3)_2$	+	" and1385(**s**) , 1370 (**m**)
$-C(CH_3)_3$	+	" and 1395 (**m**) ,1370(**s**)

Coupling of the deformation vibrations of adjacent methyl groups → the 1380 peak splits into a doublet; the splitting position is the "ID card" of the isopropyl and tert-butyl groups.

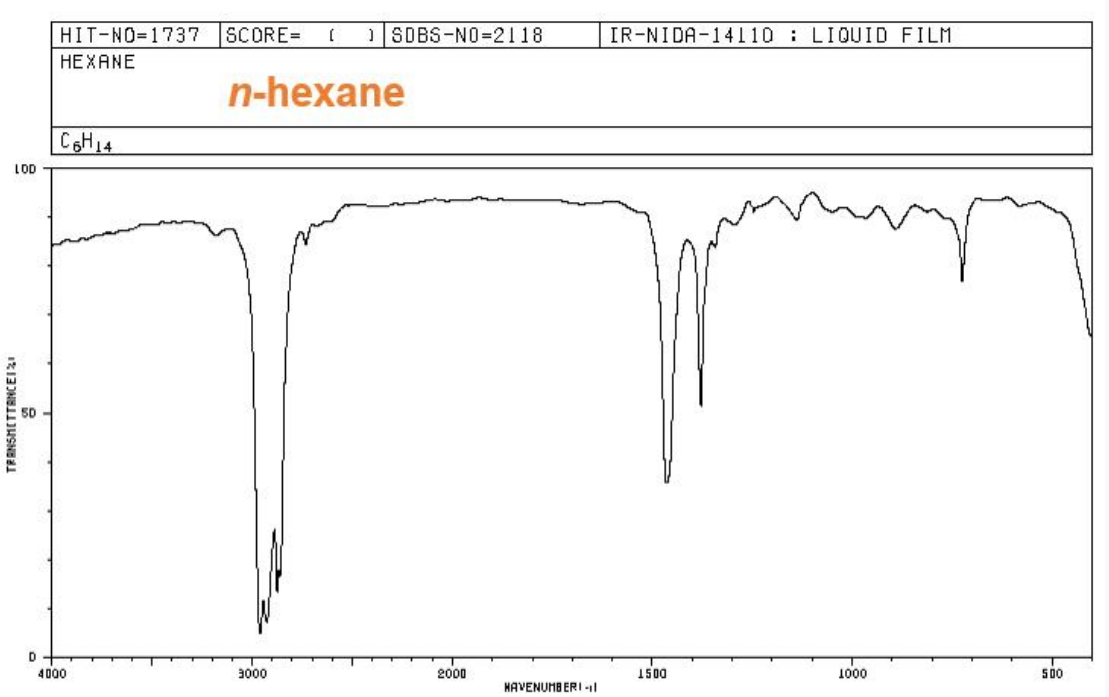
Alkanes

Example of a Branched Alkane



3176	84	1464	21	1013	77
2964	4	1380	41	972	70
2933	8	1368	47	913	84
2878	9	1297	81	797	84
2733	81	1268	86	764	74
2627	86	1175	81		
2604	86	1147	77		

Isopentane: 1385 / 1370 doublet



3187	84	1466	34	726	74
3176	84	1379	49		
2959	4	1300	84		
2928	7	1294	84		
2875	13	1138	86		
2862	15	891	84		
2734	81	884	84		

$CH_3CH_2CH_2CH_2CH_2CH_3$

Alkanes

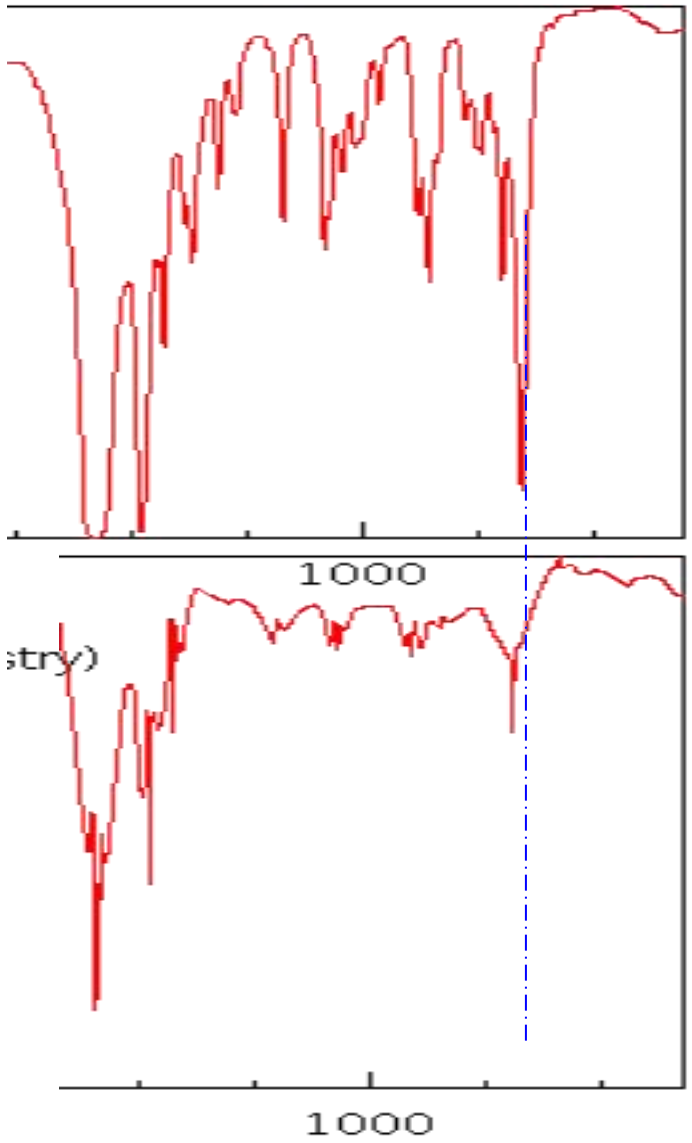
Evidence of a Long Carbon Chain: 720 cm^{-1}

$n = 1$ 770~785 cm^{-1} (medium)

$n = 2$ 740~750 cm^{-1} (medium)

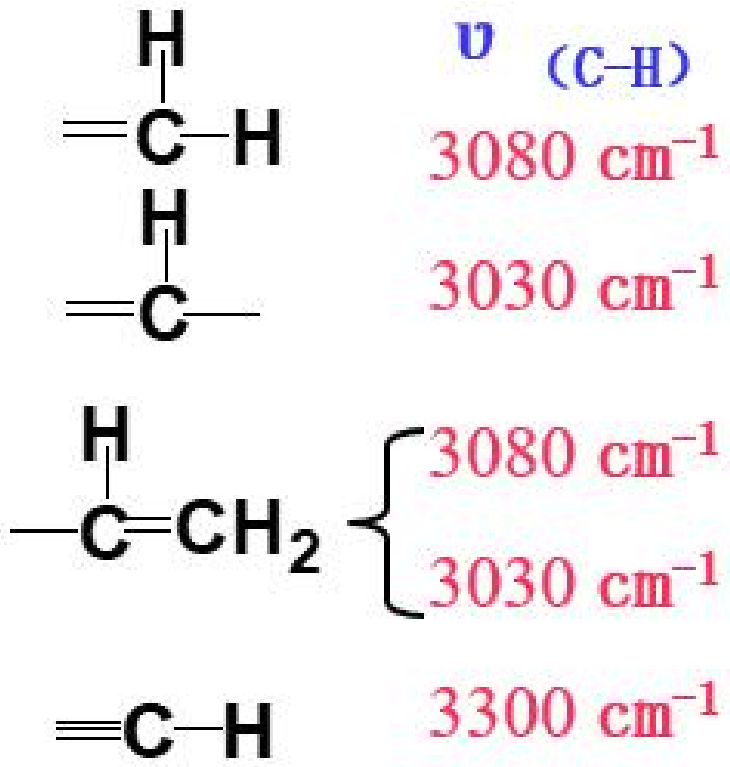
$n = 3$ 730~740 cm^{-1} (medium)

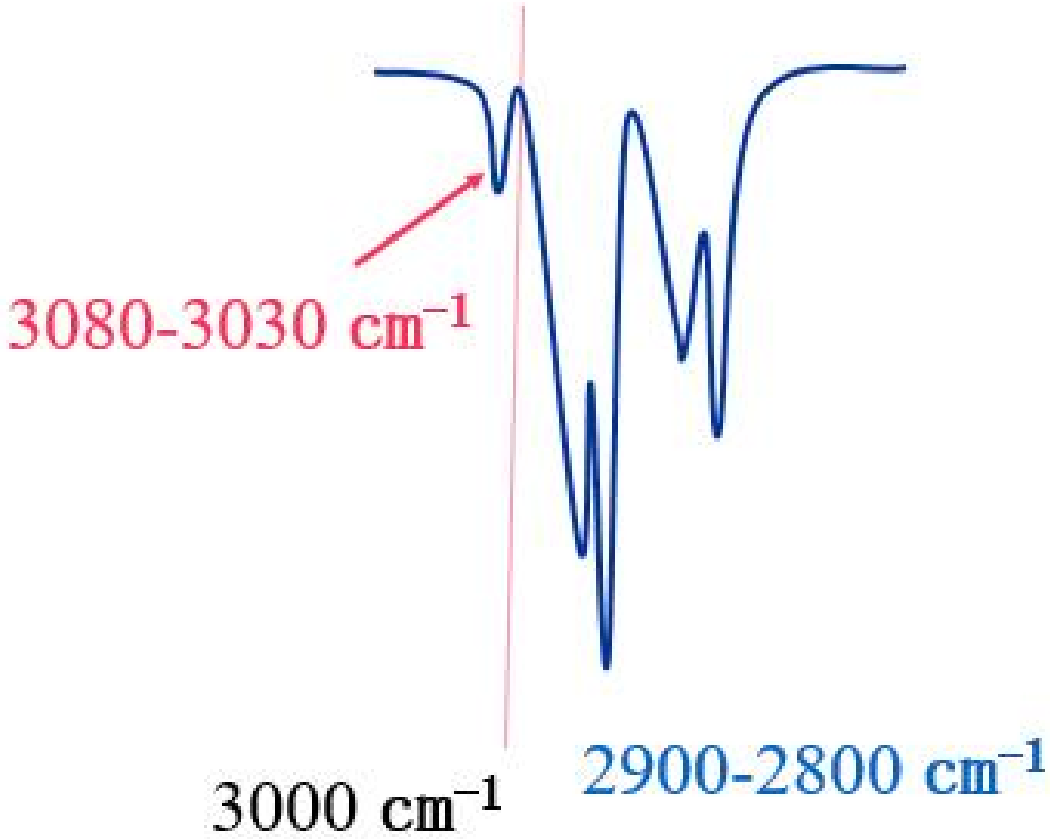
$n \geqslant$ 722 cm^{-1} (medium-strong)



Alkenes: =C—H Stretching Vibration

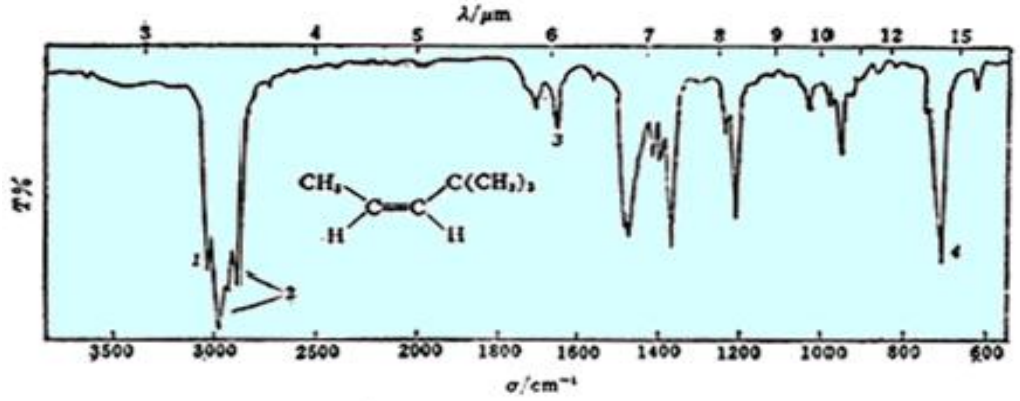
ν=C—H > 3000 cm⁻¹ (3080~3030)



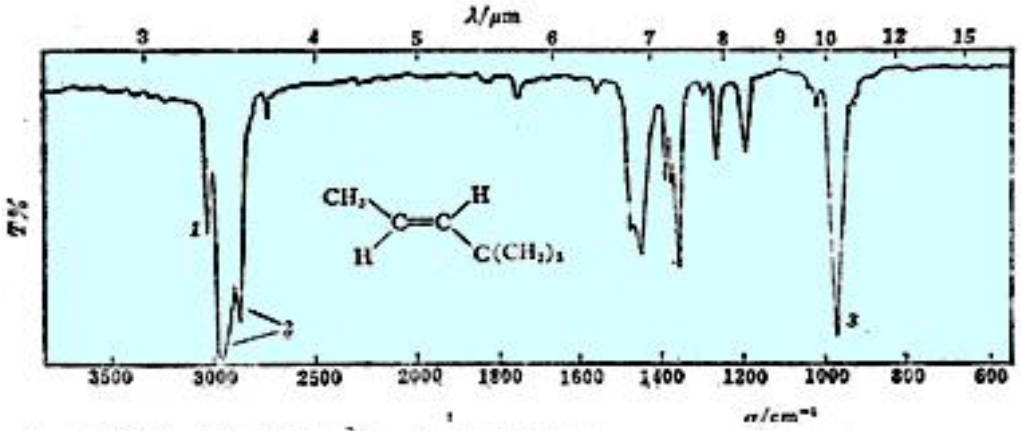


Alkenes: C=C Stretching Vibration

νC=C 1680~1630 cm^{-1} (medium-strong)



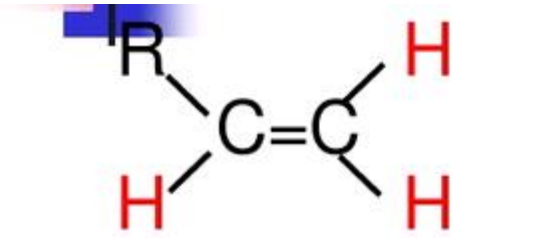
1. ≡C–H stretch; 2. =C–H stretch; 3. C≡C stretch; 4. =C–H rocking



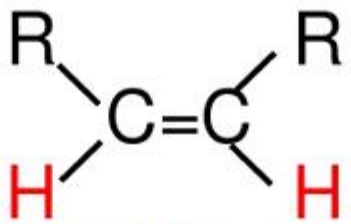
Alkenes: =C—H Bending Vibration

in-plane bending (=C-H) 1400-1420 cm^{-1} (weak)

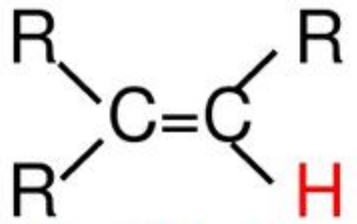
out-of-plane bending (=C-H) 1010-700 cm^{-1} (diagnostically valuable)



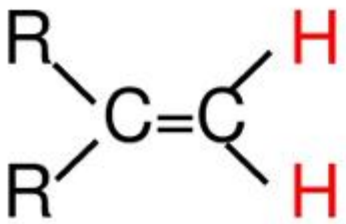
910cm^{-1}, 990 cm^{-1}



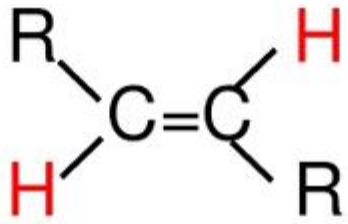
725~675cm^{-1}



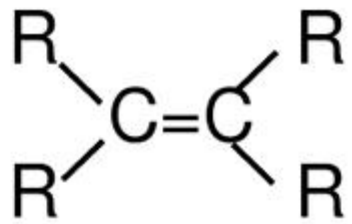
840~790cm^{-1}



890 cm^{-1}

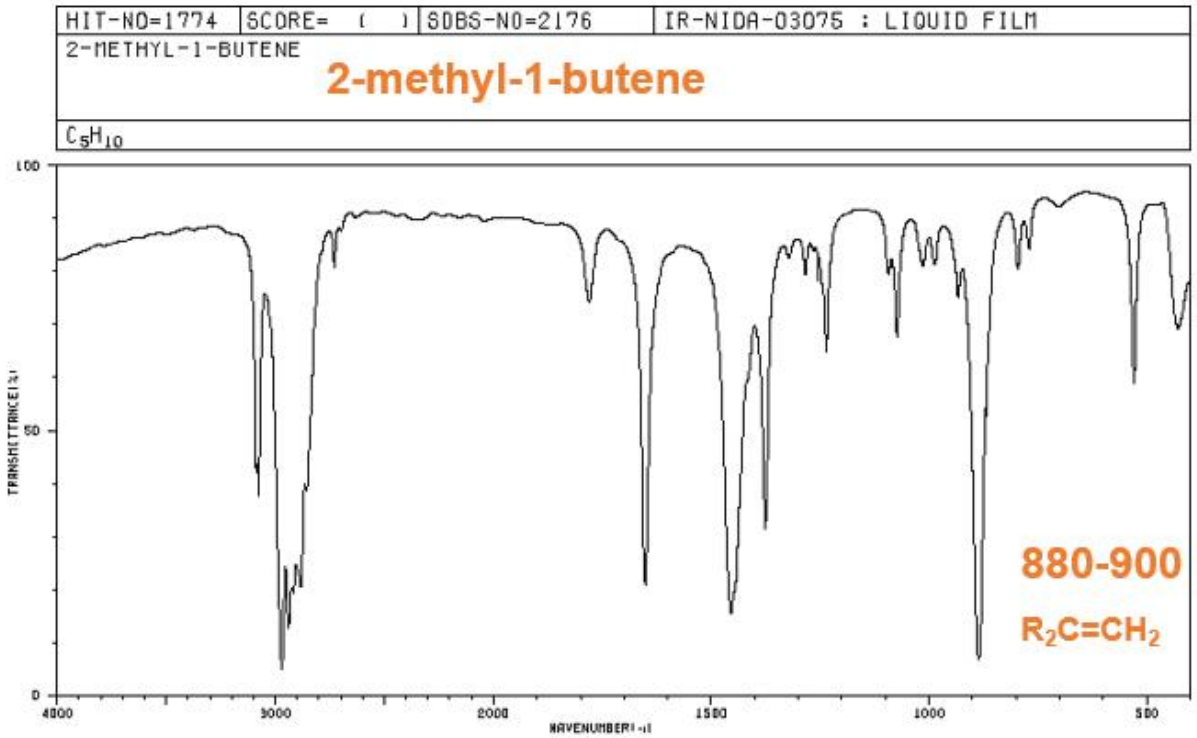


970 cm^{-1}

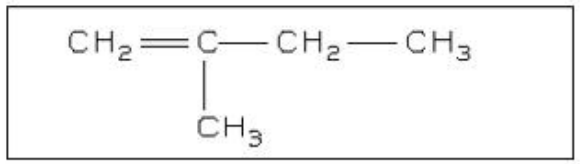


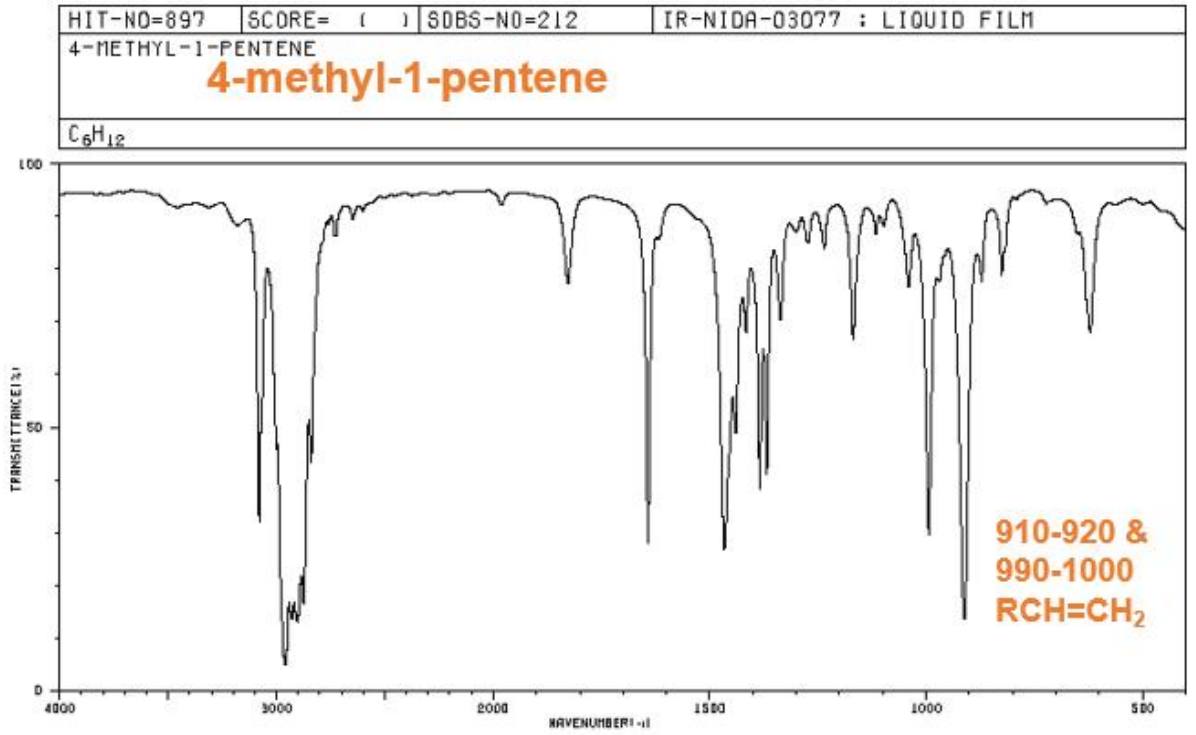
none

Alkenes Example

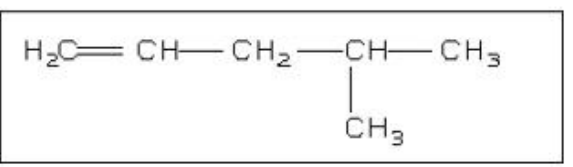


3087	41	2728	77	1266	74	887	6
3077	36	1780	72	1237	62	869	50
2970	4	1651	20	1093	77	795	77
2939	12	1466	14	1074	64	770	81
2919	18	1377	30	1014	79	530	57
2883	20	1322	79	987	79	429	66
2856	37	1284	77	933	72		





3178	84	2727	84	1386	37	1116	84	622	66		
3079	31	2648	86	1368	39	1098	84				
2960	4	1828	74	1337	68	1041	74				
2929	13	1643	28	1300	84	994	28				
2904	12	1467	26	1274	81	911	13				
2874	15	1440	47	1236	81	871	74				
2839	42	1417	66	1169	64	826	77				



Alkynes

Three Characteristic Frequencies of Alkynes

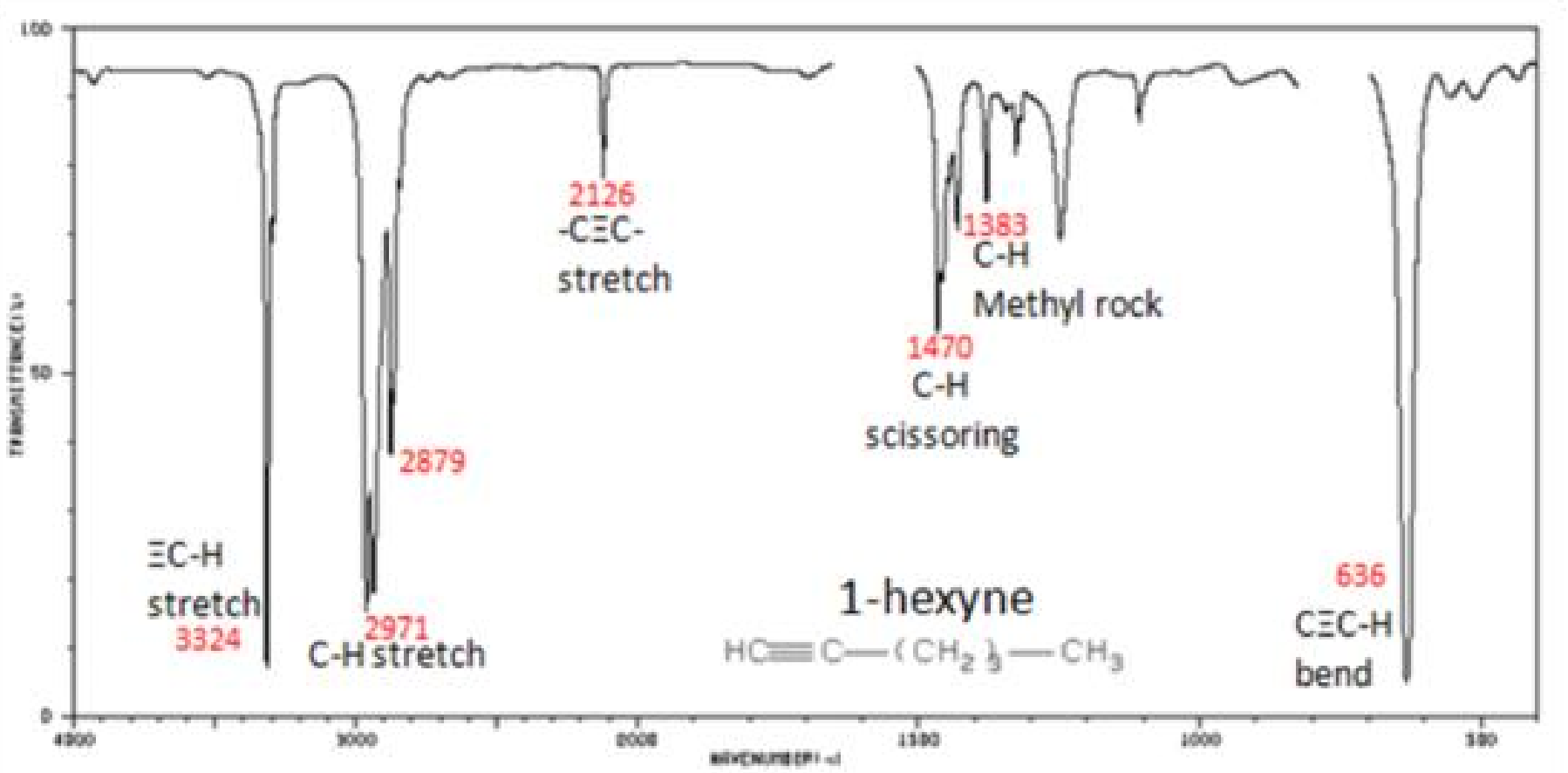
ν≡C—H | νC≡C | γC—H — the three signatures of an alkyne

Vibration	ν≡C—H	νC≡C	γC—H (out-of-plane)
σ / cm^{-1}	3333~3267	2260~2100	700~610
Features	very strong, sharp	weak (w)	strong absorption

The ≡C—H peak is narrower than both νOH and νNH, so it is easily distinguished from alcohols and amines.

Alkynes

Example of an Alkyne: 1-Hexyne



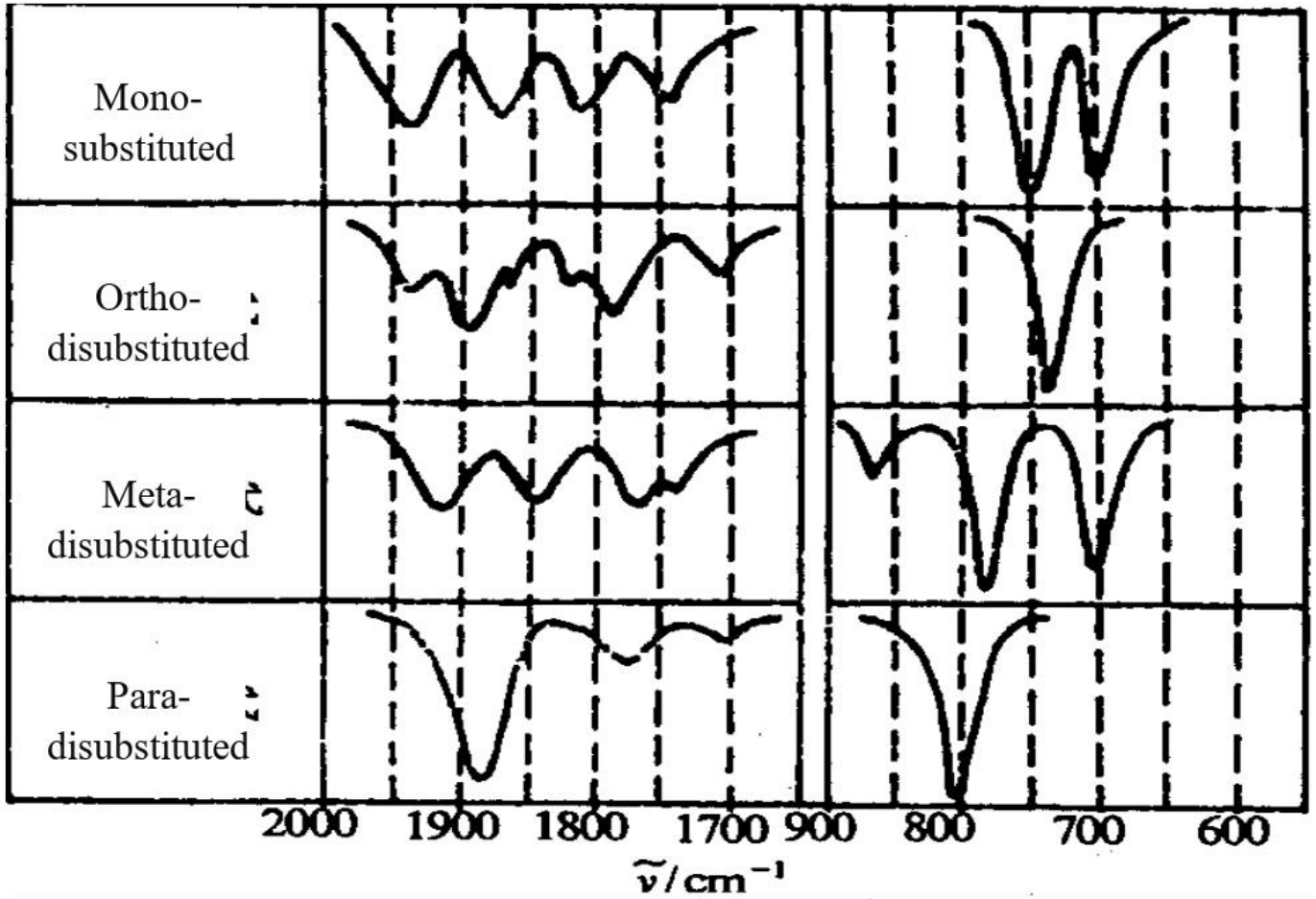
Aromatics

Four Characteristic Vibrations of Aromatic Hydrocarbons

Aromatic C—H stretch	ν=C—H 3100~3030 (m, often several peaks).
Aromatic ring skeleton	1650~1450 doublet (1600 weak, 1500 strong) — confirms the benzene ring.
Aromatic C—H out-of-plane	γ=C—H 910~665 (s) — determines the substitution pattern.
Overtone peaks	2000~1667 (vw), assisting the determination of the substitution pattern.

Aromatics

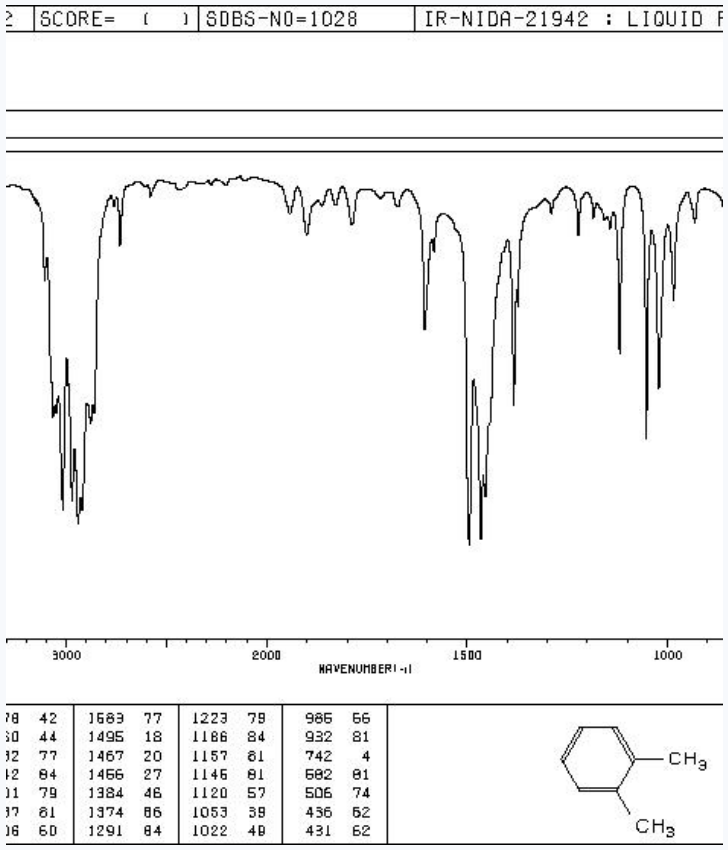
Determining the Substitution Pattern of the Aromatic Ring



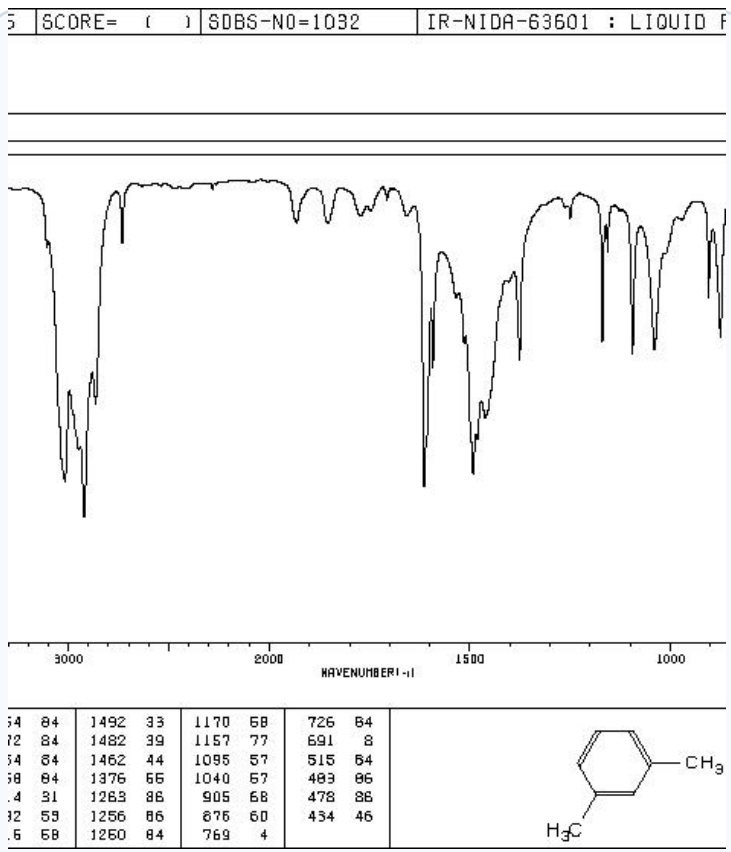
Judge together with the overtone peaks.

Aromatics

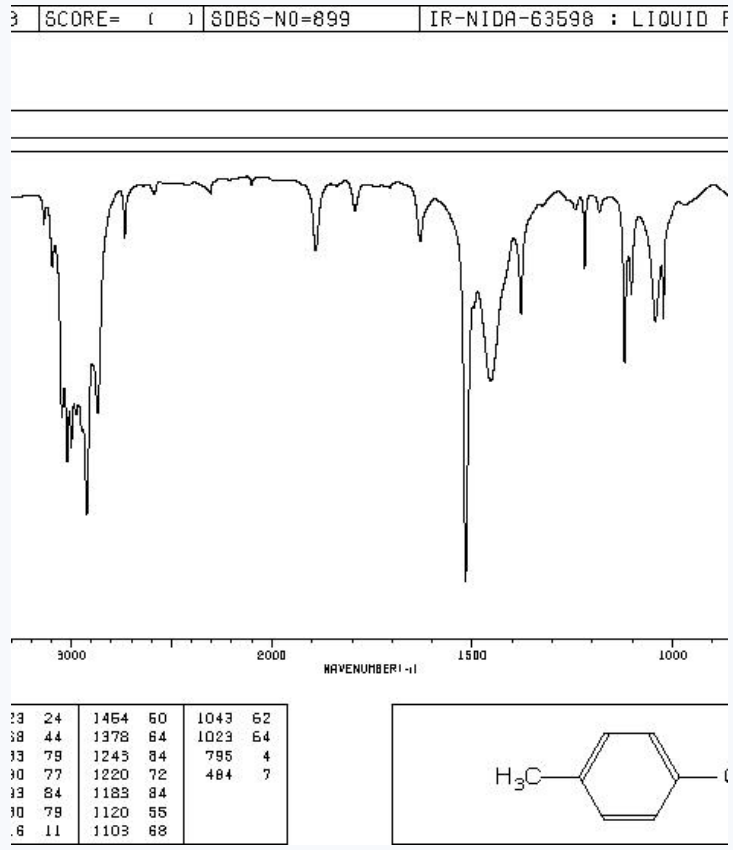
Comparison of the Three Xylene Isomers



ortho: 735-770



meta: 690-710, 750-810



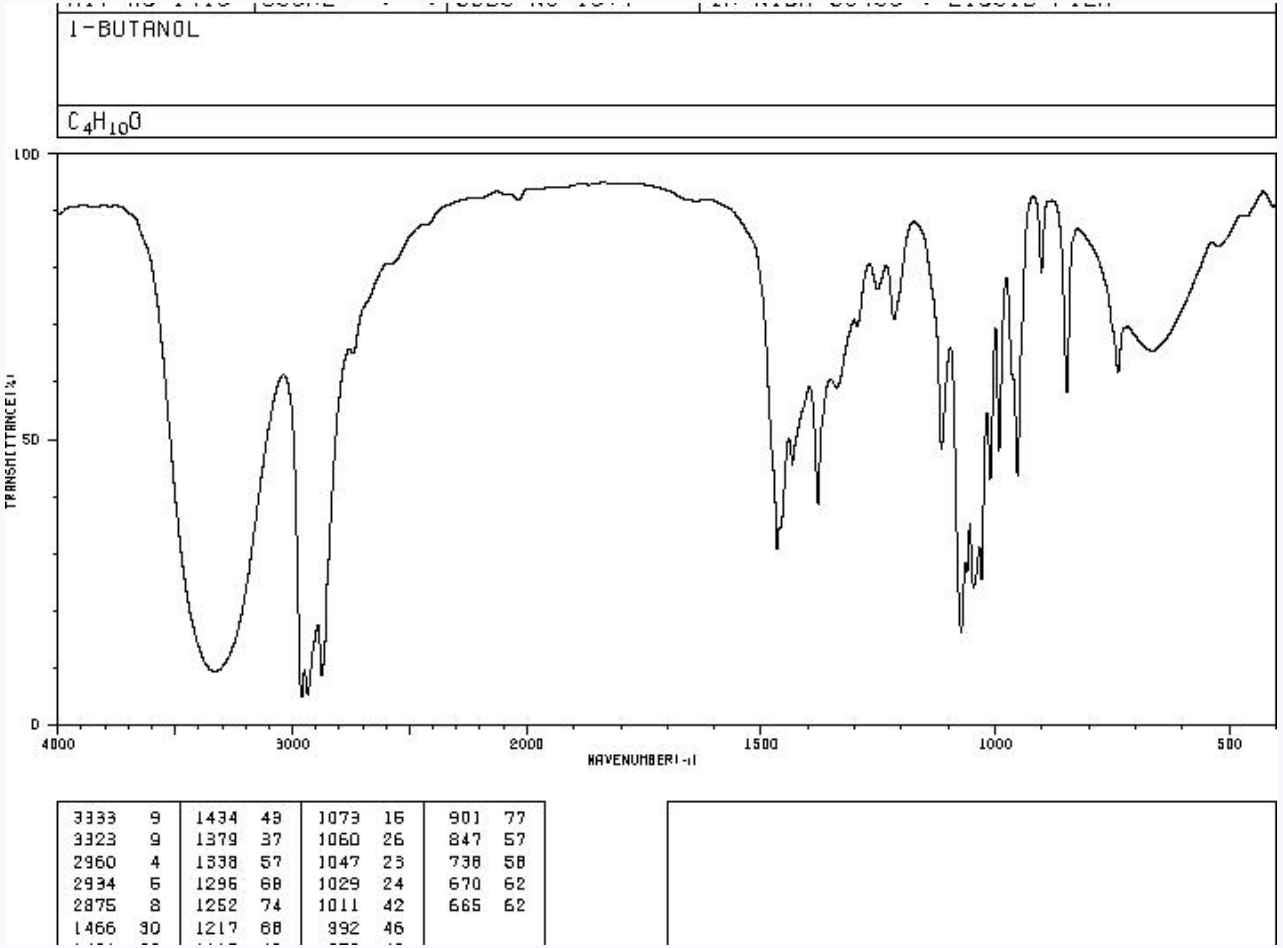
para: 810-840

Overview of Characteristic Frequencies of Alcohols and Ethers

Bond / group	σ / cm^{-1}	Remarks
O—H	**3200~3640 (broad)**	the marker peak of alcohols and phenols
C—O (primary alcohol, 1°)	1050~1085	strong absorption
C—O (secondary alcohol, 2°)	1087~1124	strong absorption
C—O (tertiary alcohol, 3°)	1124~1205	strong absorption
ether C—O—C	**1070~1150**	no O—H peak

Rule: the C—O peak of an alcohol shifts to higher wavenumber in the order 1° → 2° → 3°; the βO—H band at 1420~1330 is of poor diagnostic value.

Example of a Primary Alcohol: 1-Butanol



O—H 3200-3640 (broad)

associated hydroxyl, strong and broad.

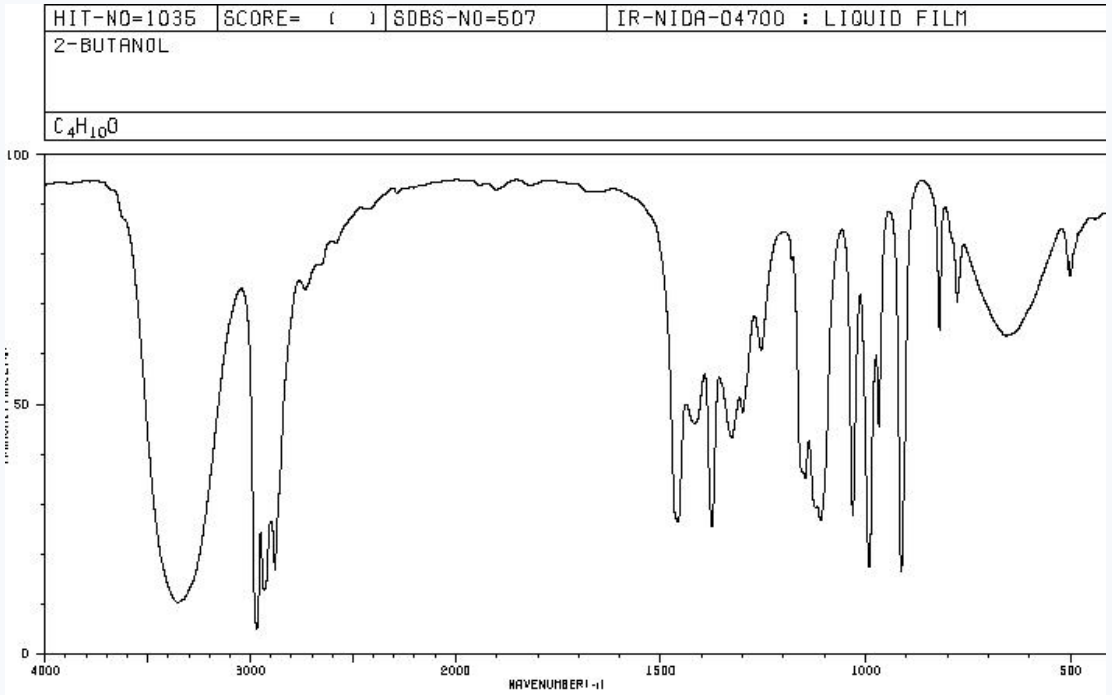
C—O 1050—1085

the characteristic position of a 1° alcohol.

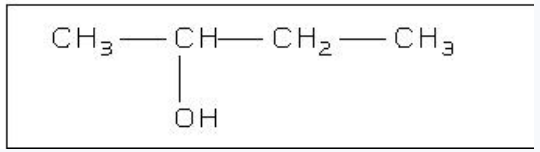
saturated C—H

below 2960, the butyl absorption.

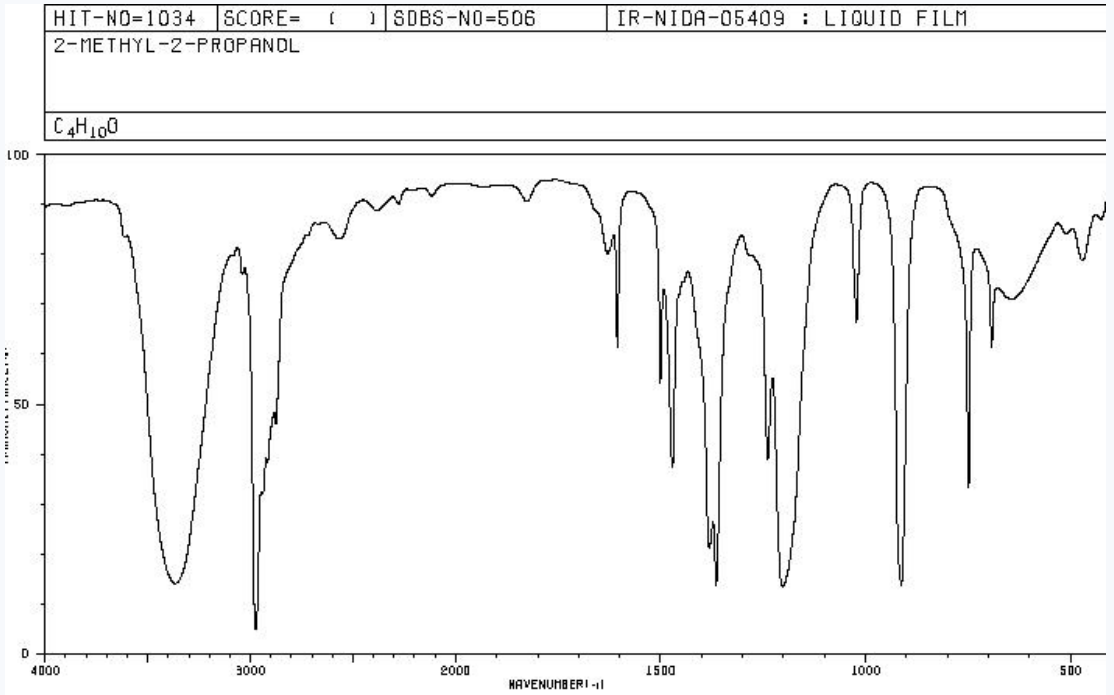
Examples of Secondary and Tertiary Alcohols



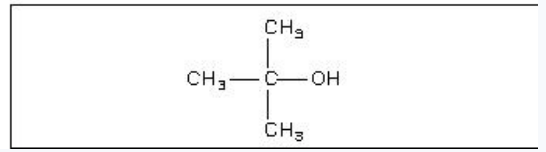
3363	10	1376	24	1110	25	666	60
2968	4	1327	42	1031	26	651	60
2932	12	1300	46	991	16	501	72
2880	16	1266	58	968	43		
2734	70	1154	35	913	15		
1457	25	1148	34	820	62		
1416	44	1122	28	777	68		



2-butanol: C—O 1087-1124 (2°)

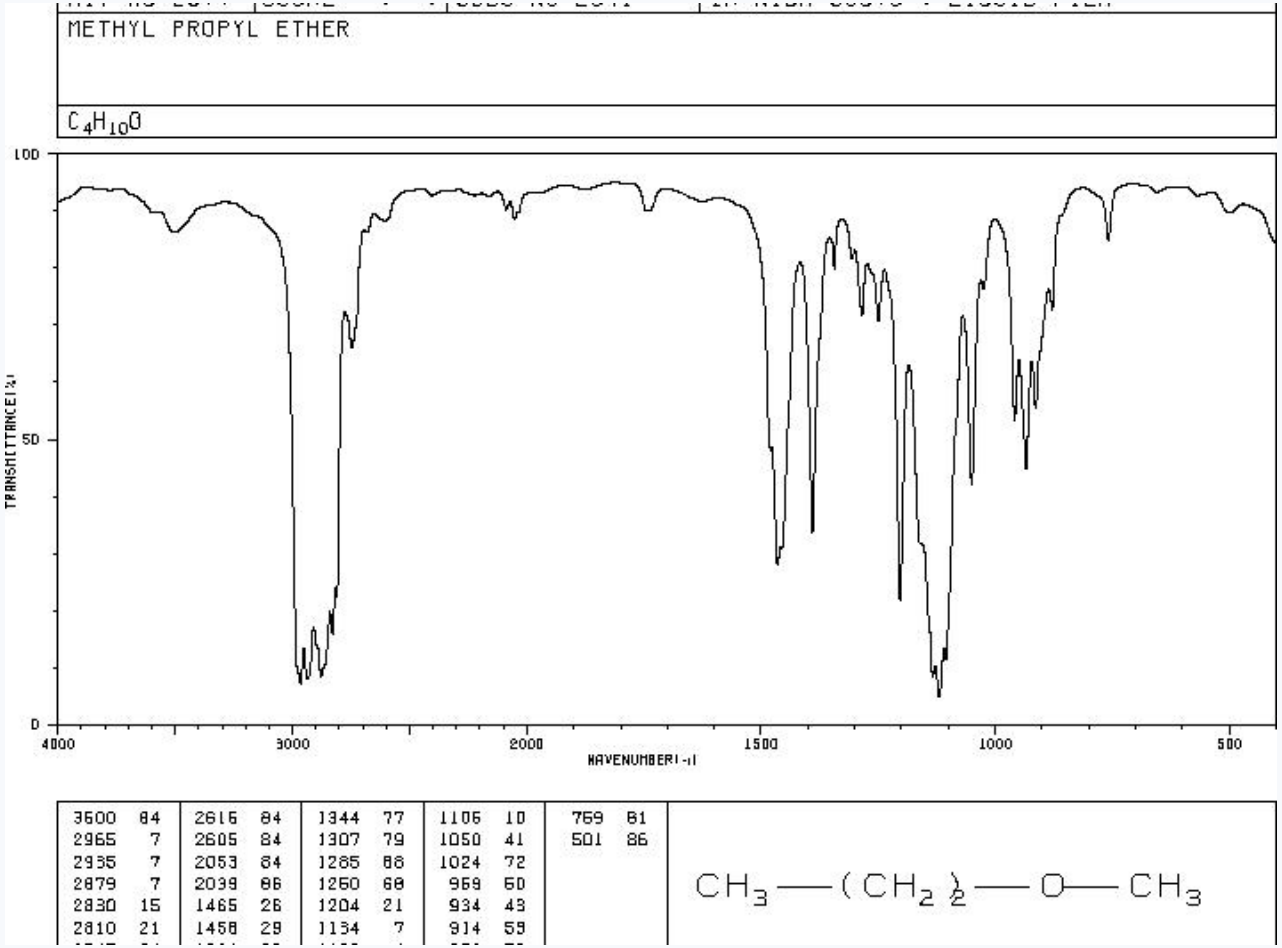


3366	13	1630	77	1202	12	611	81
3038	72	1606	58	1022	64	471	77
2974	4	1501	52	915	13		
2940	31	1471	36	749	32		
2875	44	1381	20	693	58		
2564	79	1365	19	646	66		
2385	86	1239	37	643	68		



tert-butanol: C—O 1124-1205 (3°)

Ethers: C—O Present, O—H Absent



no O—H peak

no broad peak at 3200-3640.

C—O—C 1070—1150

the characteristic strong peak of an ether.

Difference from alcohols

determined simply by the presence or absence of O—H.

Carbonyl Compounds The Carbonyl: the Leading Role of the Infrared Spectrum

The most important and most easily recognized absorption peak of the infrared spectrum — νC=O



Carbonyl Compounds

Ketones: the Reference Value of νC=O

saturated aliphatic ketone νC=O ≈ 1715 cm^{-1} (vs)

Overtone peak

the overtone of 1715 appears near 3430 (weak).

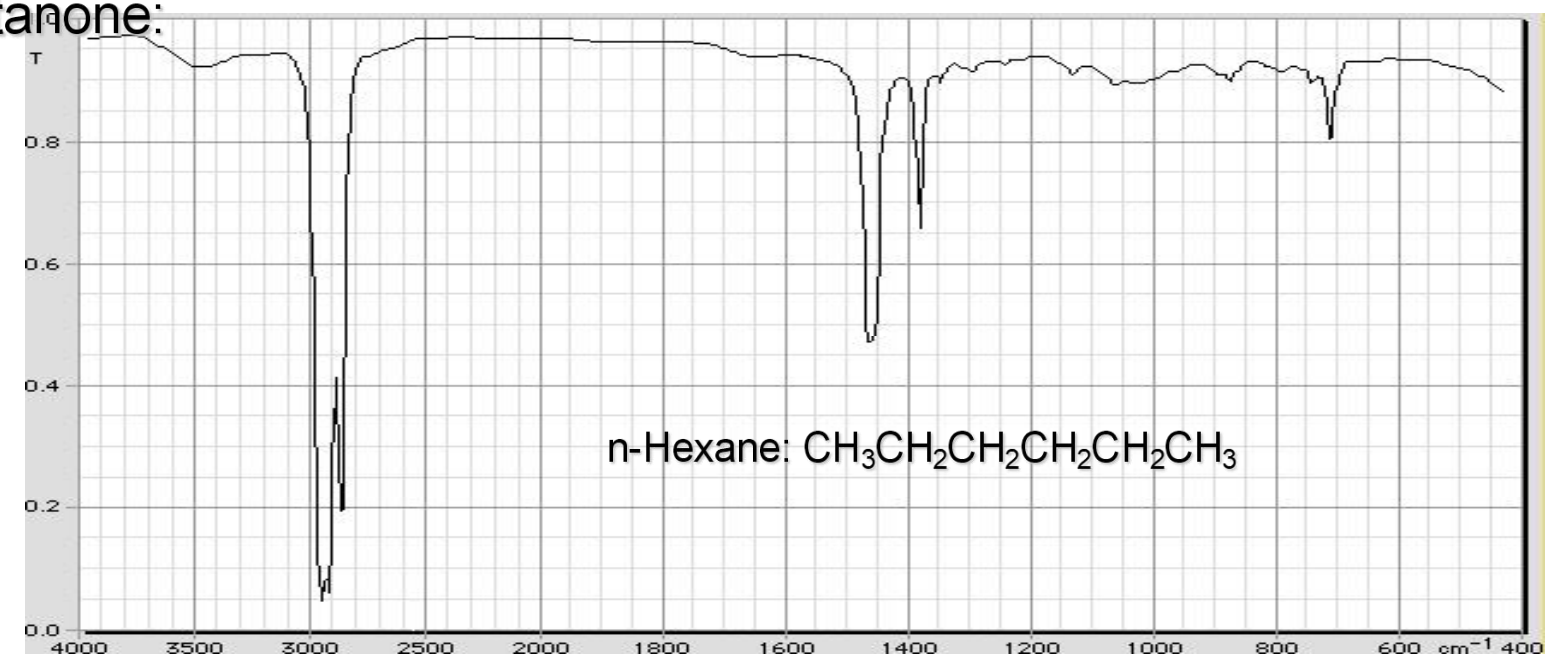
conjugation → shifts down

when conjugated with C=C or a benzene ring it shifts to lower wavenumber.

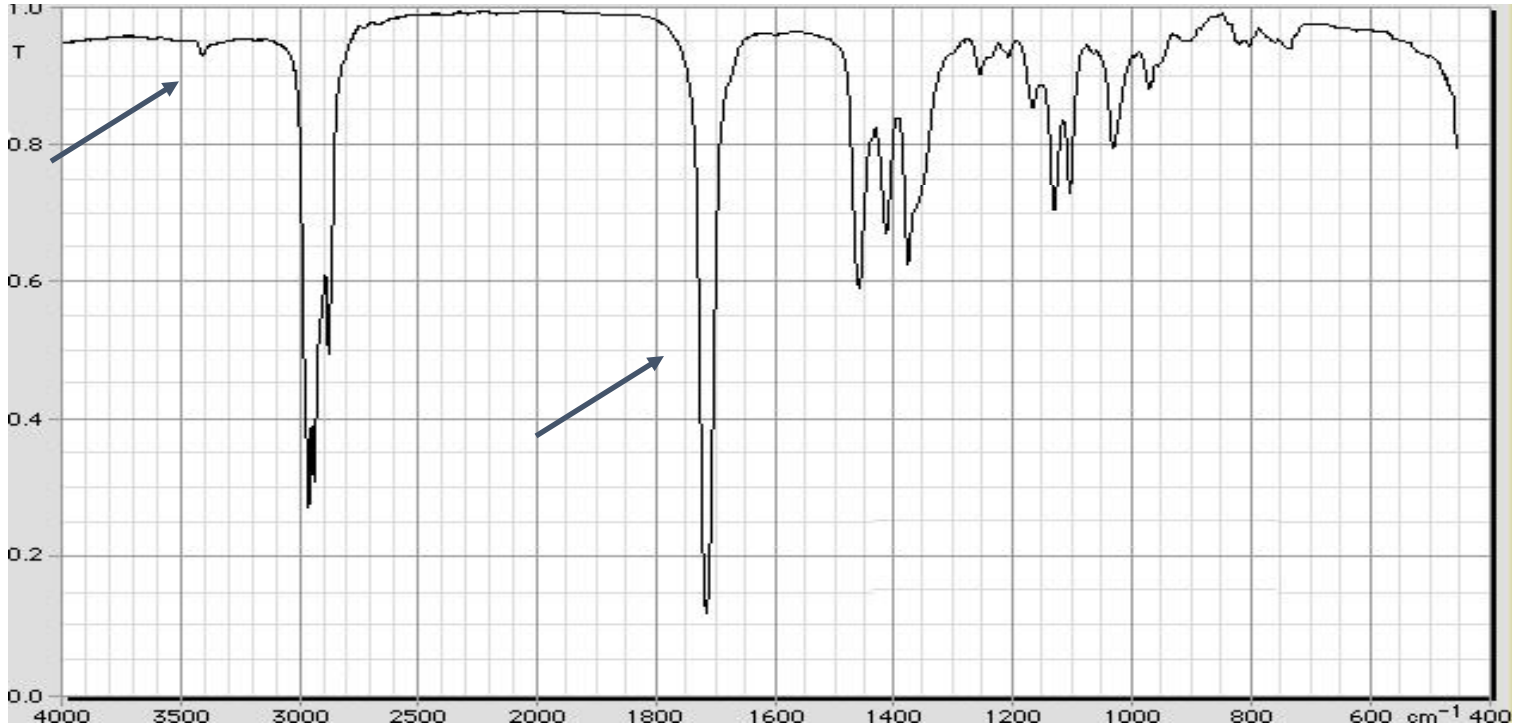
Use

1715 is the reference for judging the positions of other carbonyl peaks.

3-heptanone:



n-Hexane: $CH_3CH_2CH_2CH_2CH_2CH_3$



Carbonyl Compounds Aldehydes: the "Doublet" at 2820 and 2720

Phenomenon

the aldehyde C—H shows two medium-strong peaks at 2820 and 2720.

at the same time νC=O appears near 1725.

the doublet is the most reliable marker for recognizing aldehydes.

Mechanism

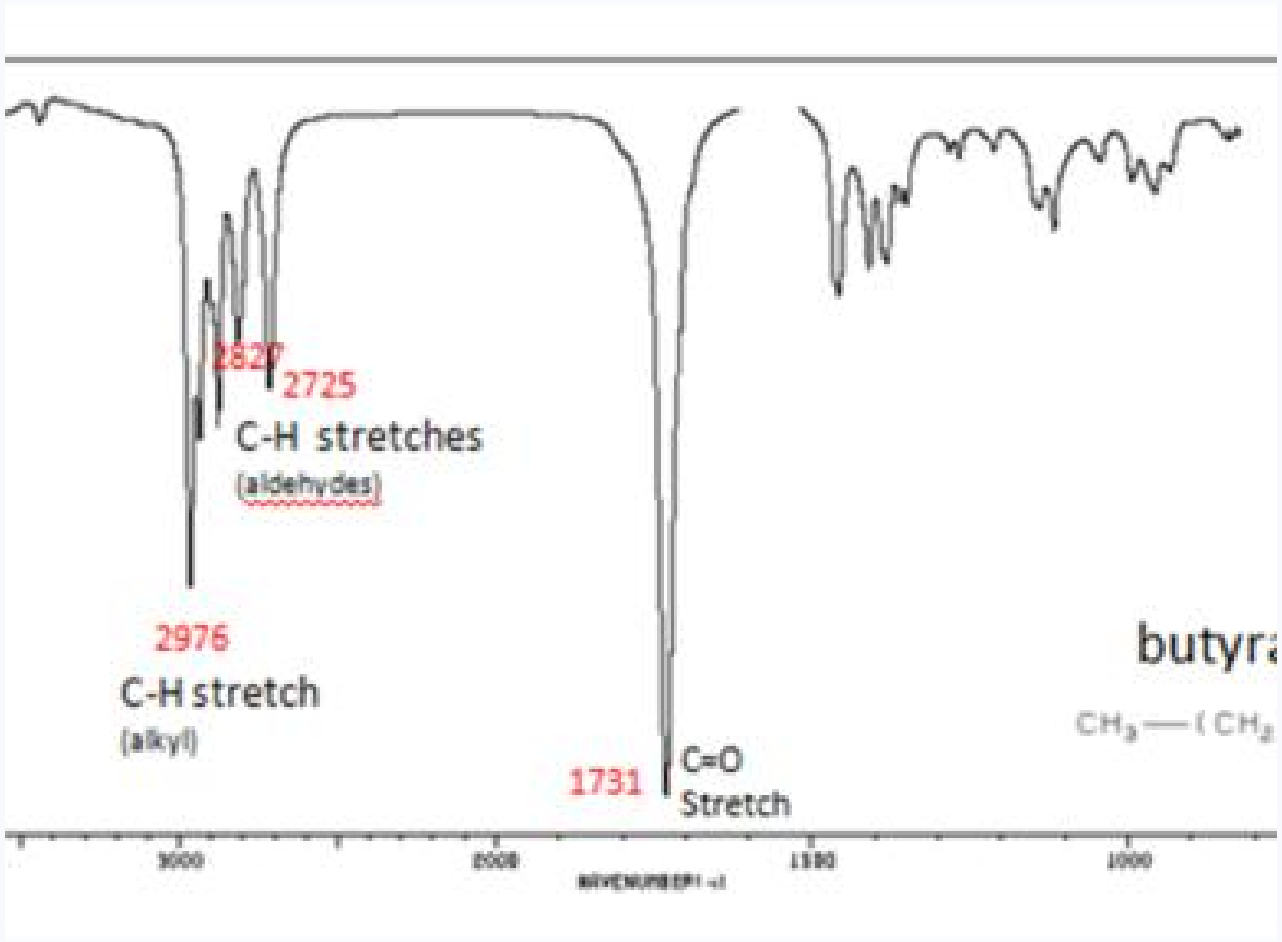
νC—H (aldehyde) ≈ 2850,

is close to the overtone of δC—H at 1390, ≈ 2780.

Fermi resonance occurs → splitting into 2820 and 2720.

Note: it is not that the aldehyde has two kinds of hydrogen; rather, Fermi resonance "splits" one peak into two.

Example of an Aldehyde: the IR Spectrum of Butanal



νC=O ≈ 1725

the carbonyl peak of a saturated aldehyde.

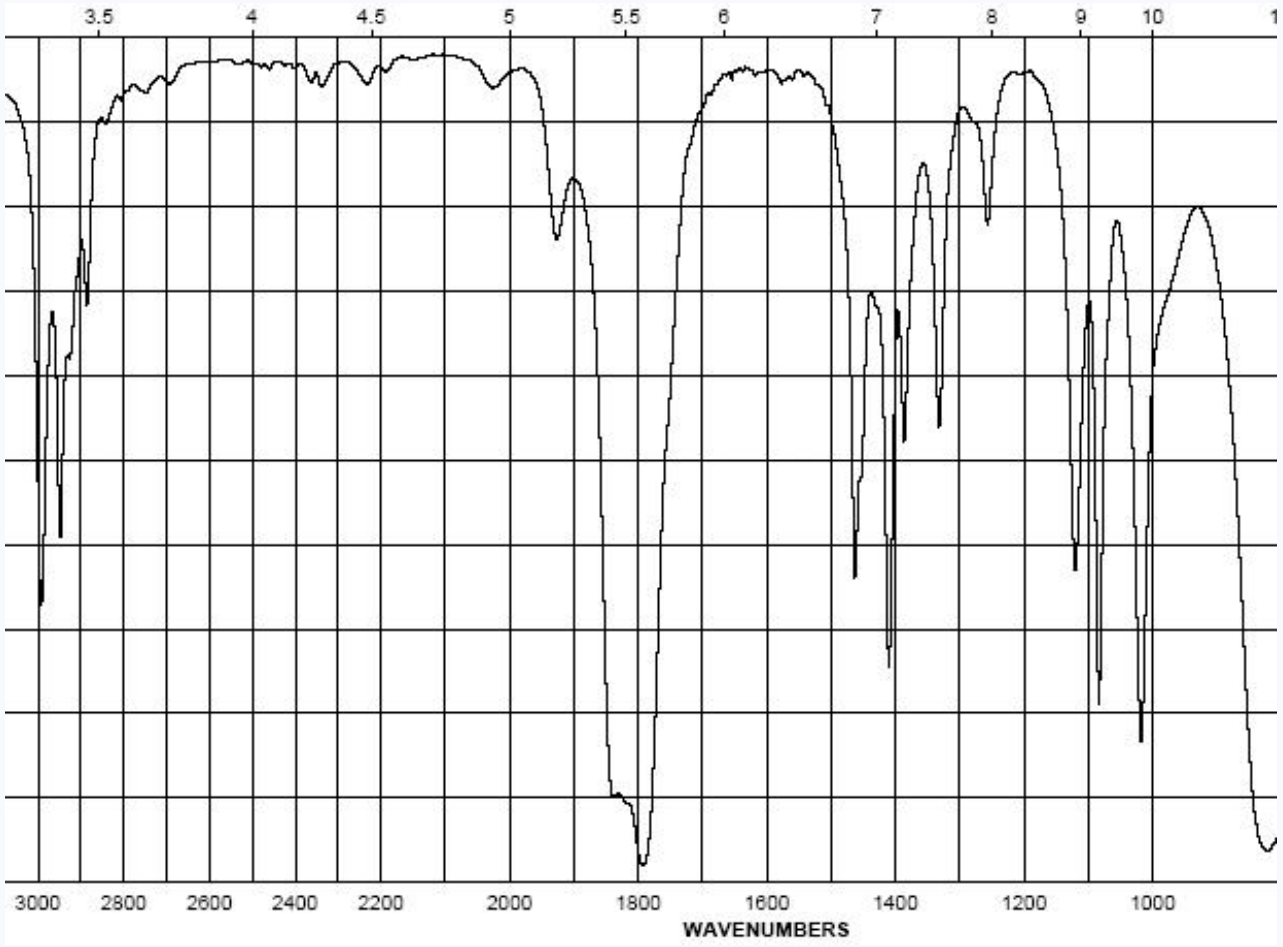
2820 and 2720

the Fermi-resonance doublet, the "ID card" of an aldehyde.

saturated C—H

below 2960, the butyl absorption.

Carbonyl Compoun Example of an Acyl Halide: Propanoyl Chloride



νC=O 1770—1820

the highest tier in the carbonyl region, strong.

C—Cl 600—800

in the fingerprint region, used as supporting evidence.

Why is it the highest?

the strong inductive effect of the halogen increases k.

Four Characteristic Peaks of Carboxylic Acids

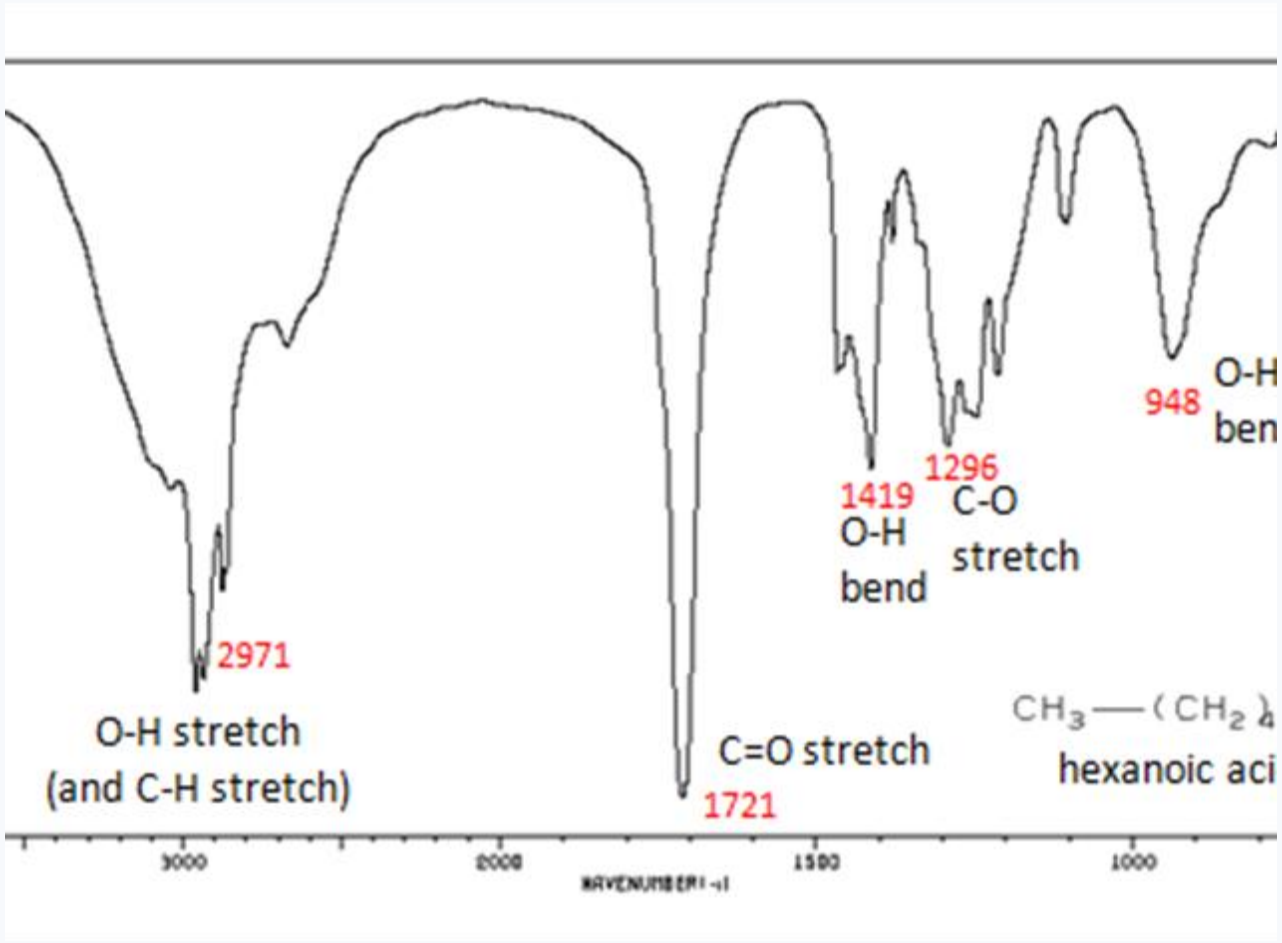
Vibration	σ / cm^{-1}	Features
νO—H	**3400~2500**	a broad peak centered on 3000 (dimer)
νC=O	**1740~1650**	strong, blunt peak
νC—O	**1320~1200**	medium-strong
γO—H	**955~915**	broad and diffuse

Monomer vs Dimer

monomer νO—H ≈ 3550 (sharp);

dimer broad and diffuse.

Example of a Carboxylic Acid: Hexanoic Acid



O—H 3400—2500

a broad peak centered on 3000.

νC=O ≈ 1710

a strong, blunt carbonyl peak.

C—O and γO—H

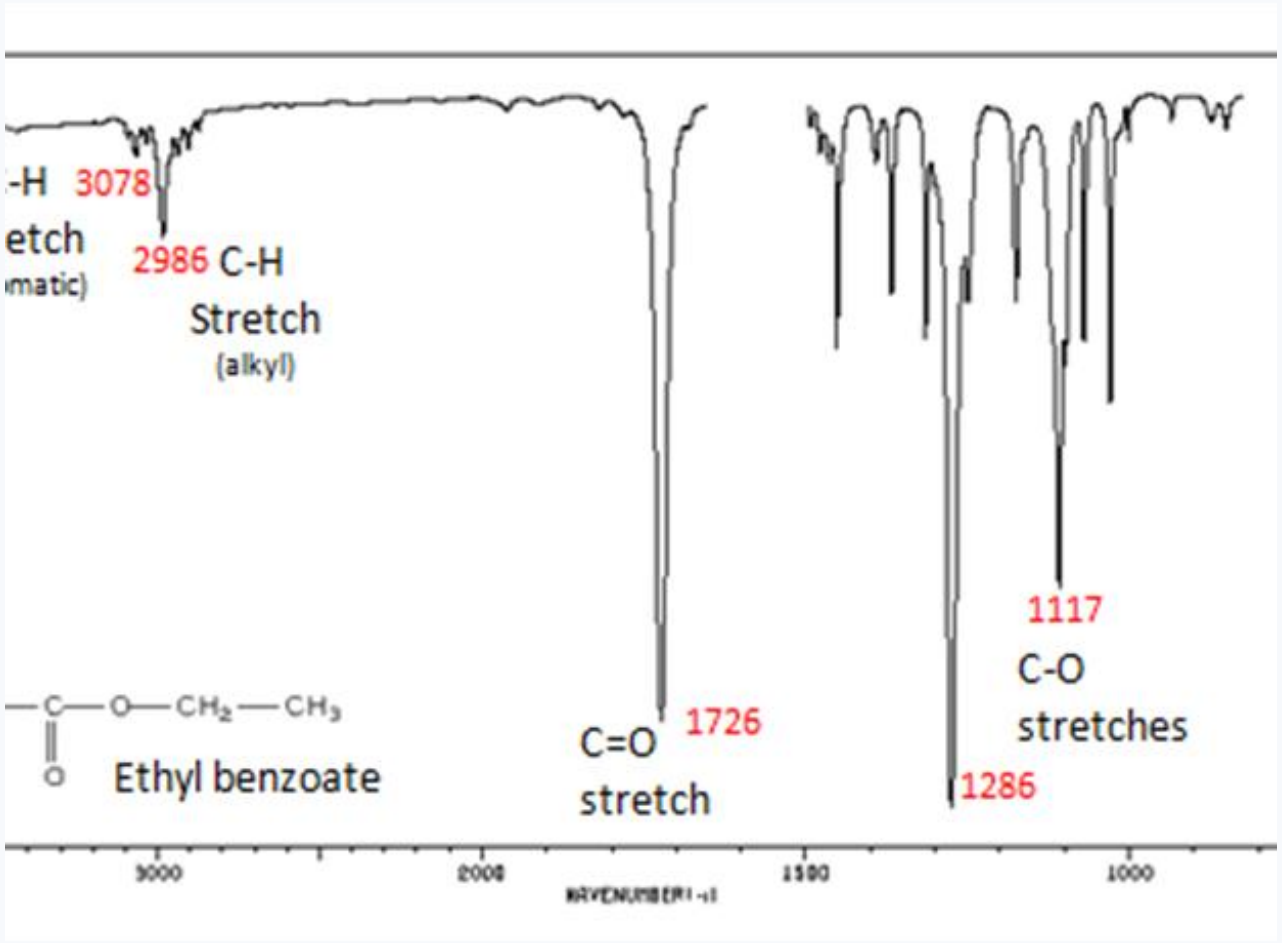
1320-1200 and 955-915.

Two Characteristic Frequencies of Esters

Vibration	σ / cm^{-1}	Features
νC=O	≈ 1735	strong, the reference value; shifts down on conjugation
νas(C—O—C)	**1300~1150**	strong and broad
νs(C—O—C)	**1150~1000**	weaker

Memory aid: ester = a strong carbonyl at 1735 + two strong C—O—C peaks at 1300~1000; sometimes a weak overtone is also seen near 3450.

Carbonyl Compoun Aromatic Ester: Ethyl Benzoate



νC=O below ≈ 1720

conjugation with the benzene ring shifts it down.

two strong C—O—C peaks

1300-1150 and 1150-1000.

the three signatures of the benzene ring

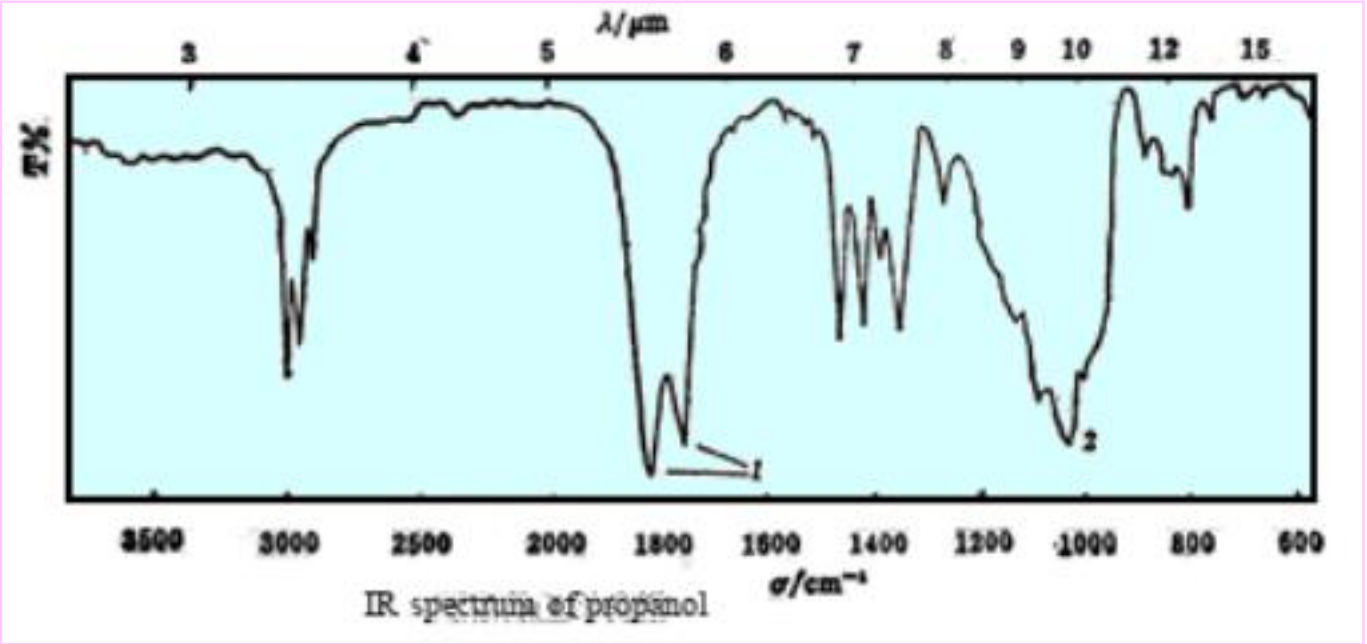
near 3030, 1600/1500 and 700.

Carbonyl Compoun Acid Anhydrides: the "Doublet" of C=O

νas 1850~1800 + νs 1780~1740 cm^{-1}

νC—O 1100~1000

strong absorption, in the fingerprint region.



IR spectrum of propanol

Amides: Three Types of Substitution

Type	νNH / cm^{-1}	Remarks
primary amide	**doublet: 3530 and 3400**	free; shifts down on association
secondary amide	**singlet: 3500~3400 (sharp)**	associated 3330~3060
tertiary amide	**no νNH peak**	no hydrogen on the nitrogen

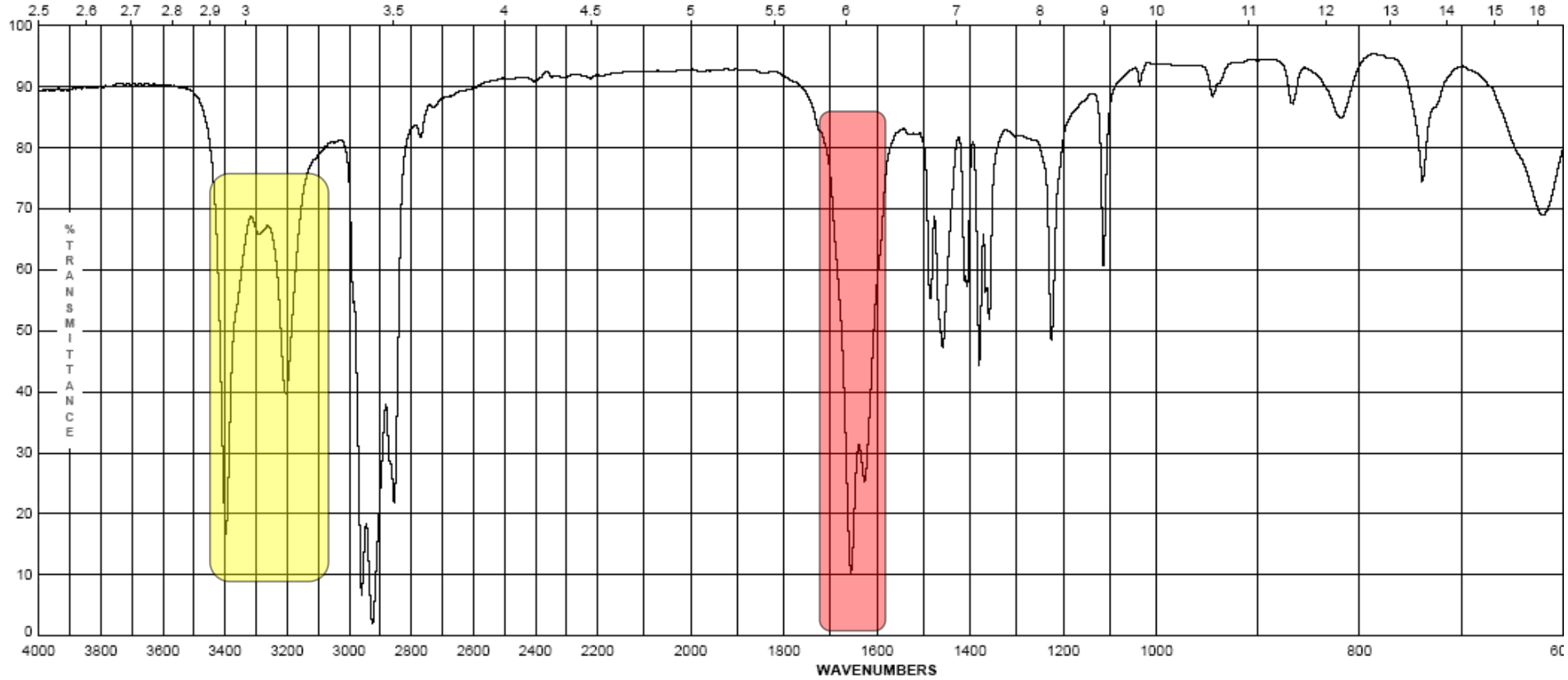
νC=O 1680~1630

p-π conjugation shifts the peak to lower wavenumber.

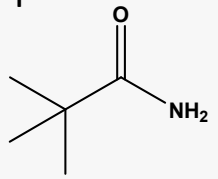
βNH 1640~1550

weaker, close to νC=O.

Example of an Amide: Palmitamide



pivalamide



Nitrogen Compounds: Characteristic Frequencies of Amines

Vibration	σ / cm^{-1}	Features
νNH	**3500~3300**	medium-strong, sharp; primary amine doublet, secondary amine singlet, tertiary amine none
δNH	**1650~1580**	m→s; stronger for aromatic amines
νC—N	**1350~1000**	medium-strong

Nitrogen Compoun Comparison of the Spectra of the Three Amines

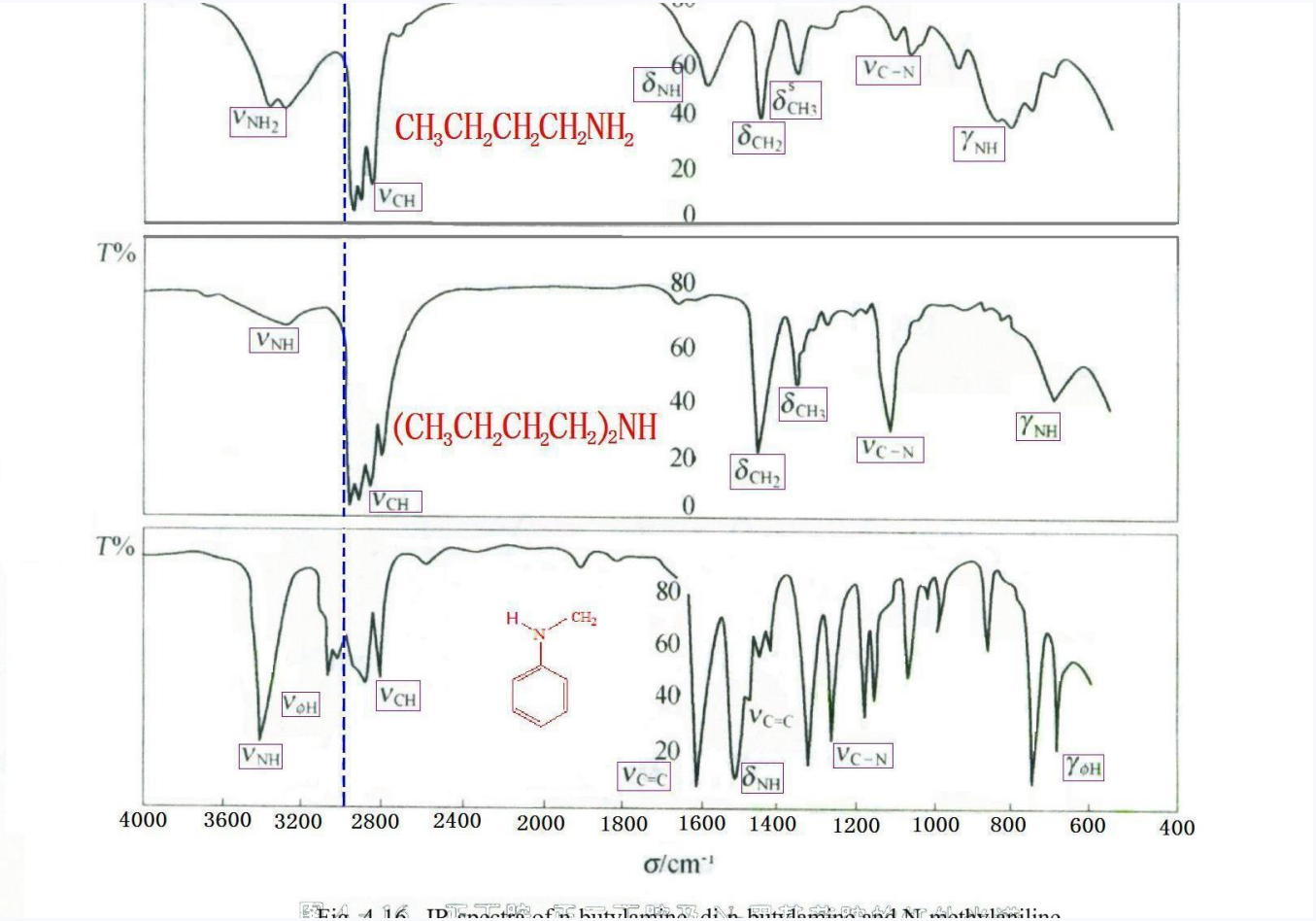


Fig. 4.16 IR spectra of n-butylamine, di-n-butylamine and N-methylaniline

primary amine 3500 / 3400

doublet.

secondary amine ≈ 3350

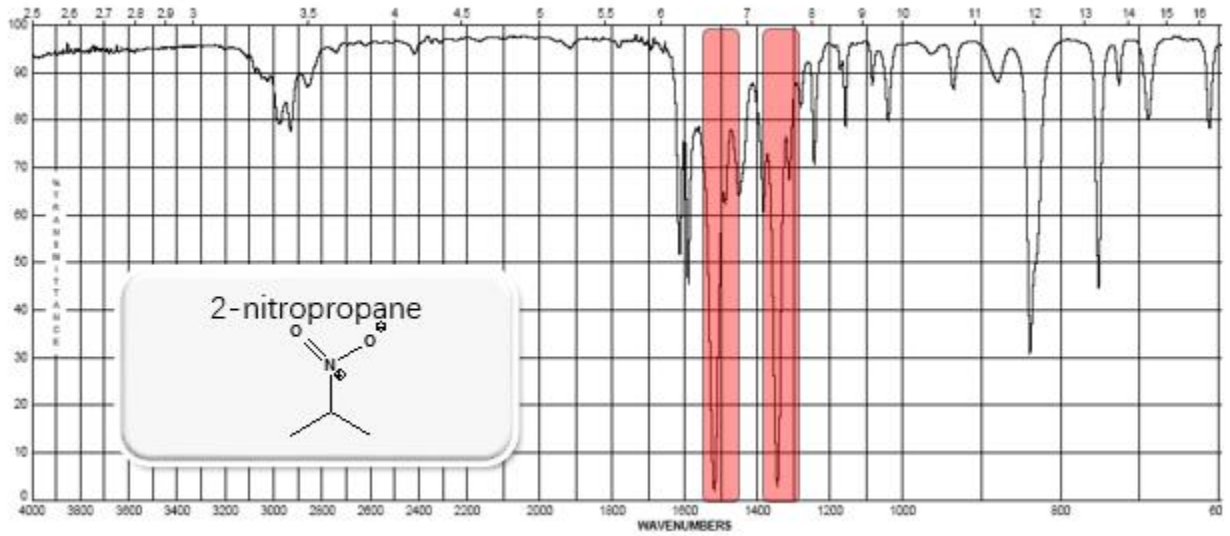
singlet.

tertiary amine

no νNH peak.

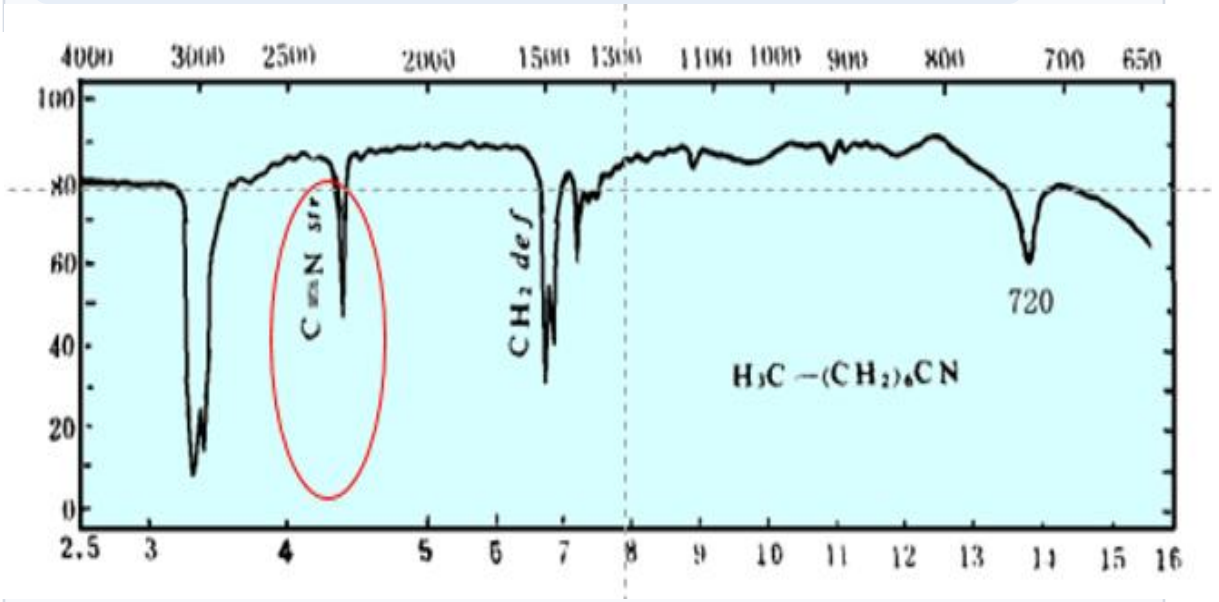
Nitrogen Compounds Nitro Compounds and Nitrile Compounds

nitro —NO_2 (two strong peaks appearing as a pair)



νas 1550~1570 (strong); νs 1350~1380 (strong).

nitrile —C≡N (characteristic, strong and sharp)



νC≡N 2260~2220 cm^{-1}. Left spectrum: $H_3C—(CH_2)_4—CN$.

04

Section 4 • IR Spectral Analysis

From Spectrum to Structure — Methods and Examples of Interpretation

• Preparation before interpretation • Degree of unsaturation • Order of interpretation • Worked examples

Preparation Before Interpretation

1. Learn the origin and properties of the sample to judge its possible class.
2. Thoroughly remove the solvent from the sample to avoid interfering peaks.
3. Measure the physicochemical constants of the sample (boiling point, refractive index, etc.).
4. Calculate the degree of unsaturation from the molecular formula to estimate the type of compound.

Calculation of the Degree of Unsaturation

$$U = n_4 + 1 + \frac{n_3 - n_1}{2}$$

U	Structural meaning
0	saturated acyclic compound
1	**one double bond or one ring**
2	**one triple bond**
4	**one benzene ring**

n_4, n_3 and n_1 are the numbers of tetravalent, trivalent and monovalent atoms.

Benzaldehyde (C_7H_6O)

$$U = \frac{2 + 2n_4 + n_3 - n_1}{2} = \frac{2 + 2 \times 7 - 6}{2} = 5$$

n-Pentane (C_5H_{12})

$$U = \frac{2 + 2 \times 5 - 12}{2} = 0$$

$$U = \frac{2 + 2n_4 + n_3 - n_1}{2}$$

Order of Spectrum Interpretation

Step 1: Group-frequency region

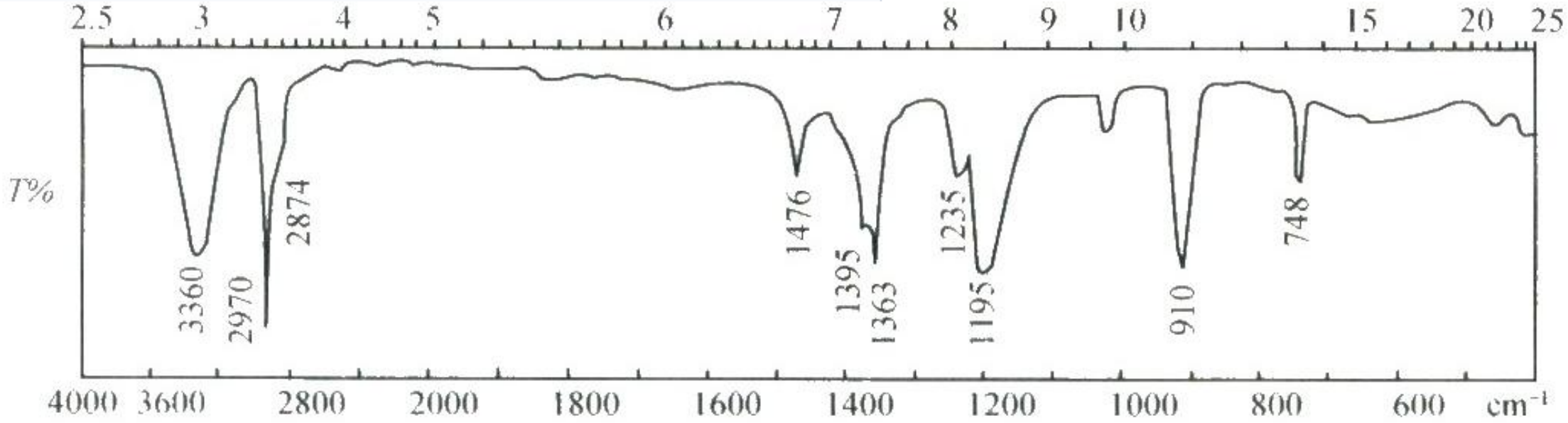
Identify functional groups, classify compound

Step 2: Fingerprint region

Use fingerprint bands to confirm structure

A known compound can be checked directly against a reference spectrum; an unknown is identified by combining the degree of unsaturation with other data.

Worked Example: $C_4H_{10}O$

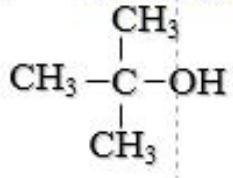


Solution: $U = (2 + 2\times4 - 10) / 2 = 0$, *saturated aliphatic compound*

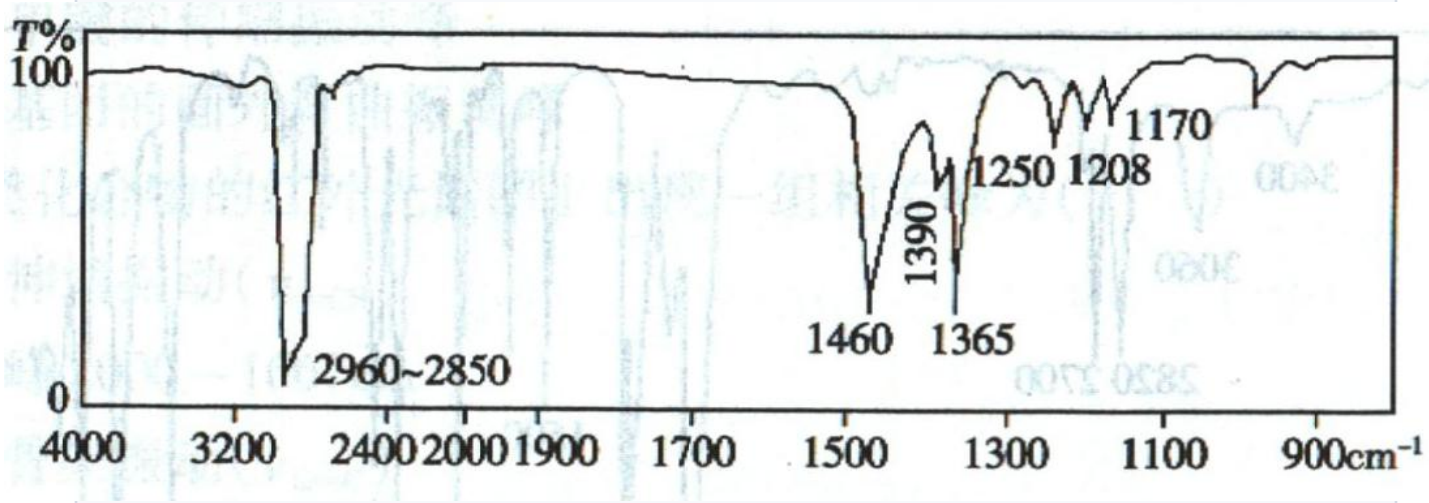
Characteristic region:

1. 2970 cm^{-1} (vs): $\nu_s(CH_2)$ of tert-butyl; $\delta_s(CH_3)$ 1395, 1363 cm^{-1} (doublet)
2. 3360 cm^{-1} (broad, s): ν_{O-H} of tert-alcohol; ν_{C-O} 1195 cm^{-1}

Structure:



Worked Example: C_8H_{18}



U = 8+1+(0−18)/2 = 0

a saturated alkane.

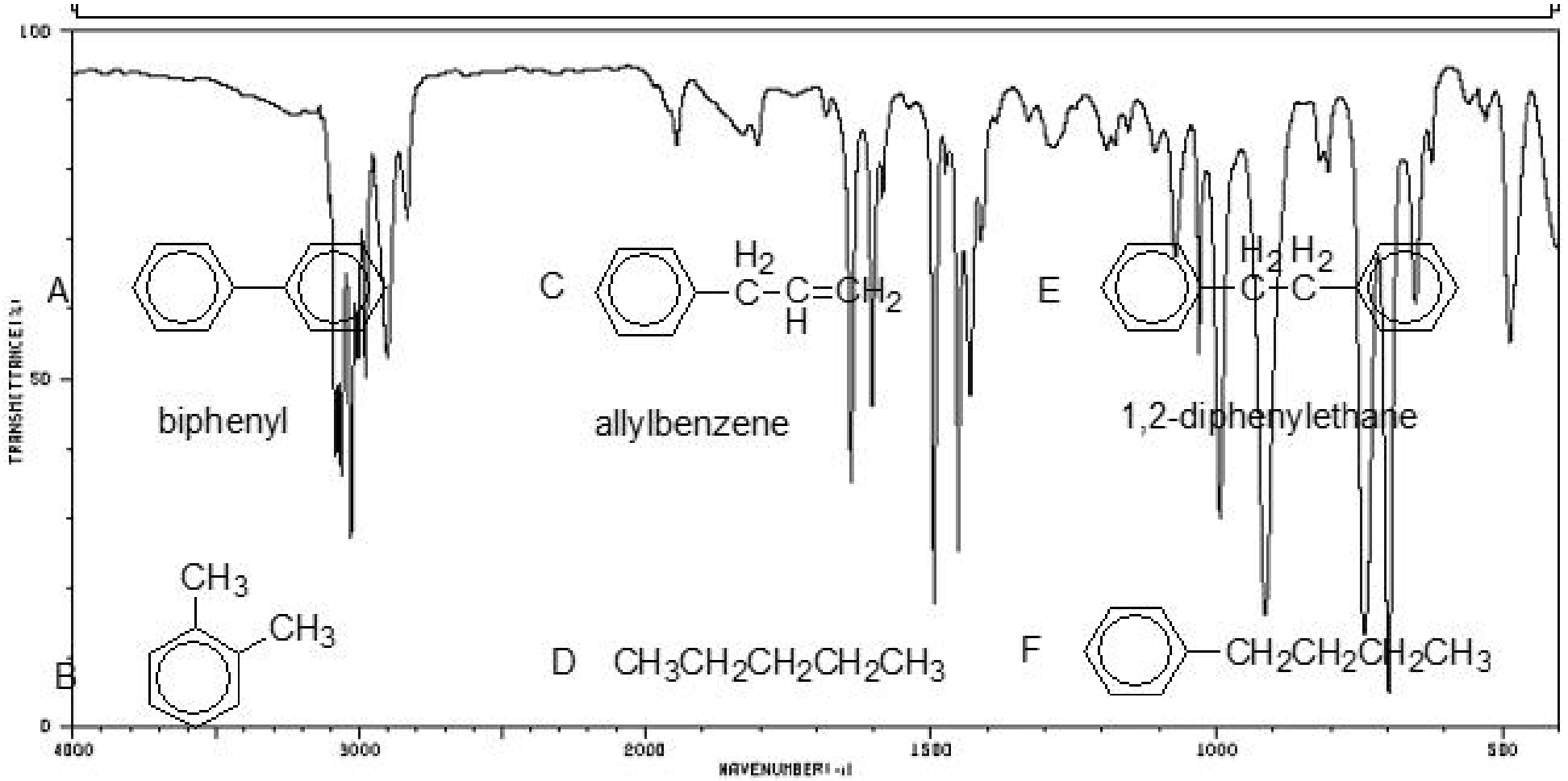
2960~2850 strong absorption

saturated C—H stretch.

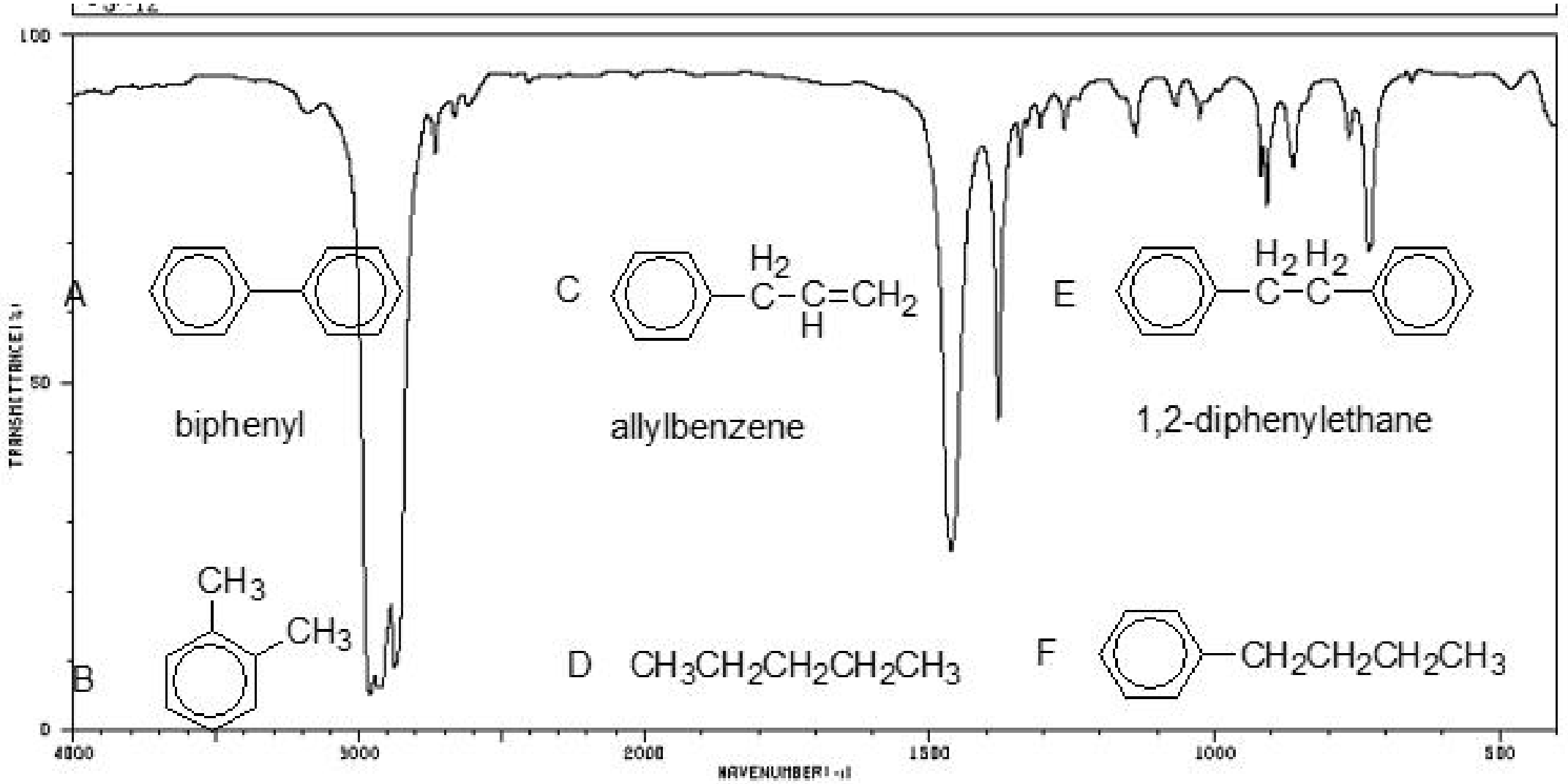
1460 and 1390/1365 doublet

CH_2 /CH_3 bending vibrations; the doublet indicates branching (isooctane type).

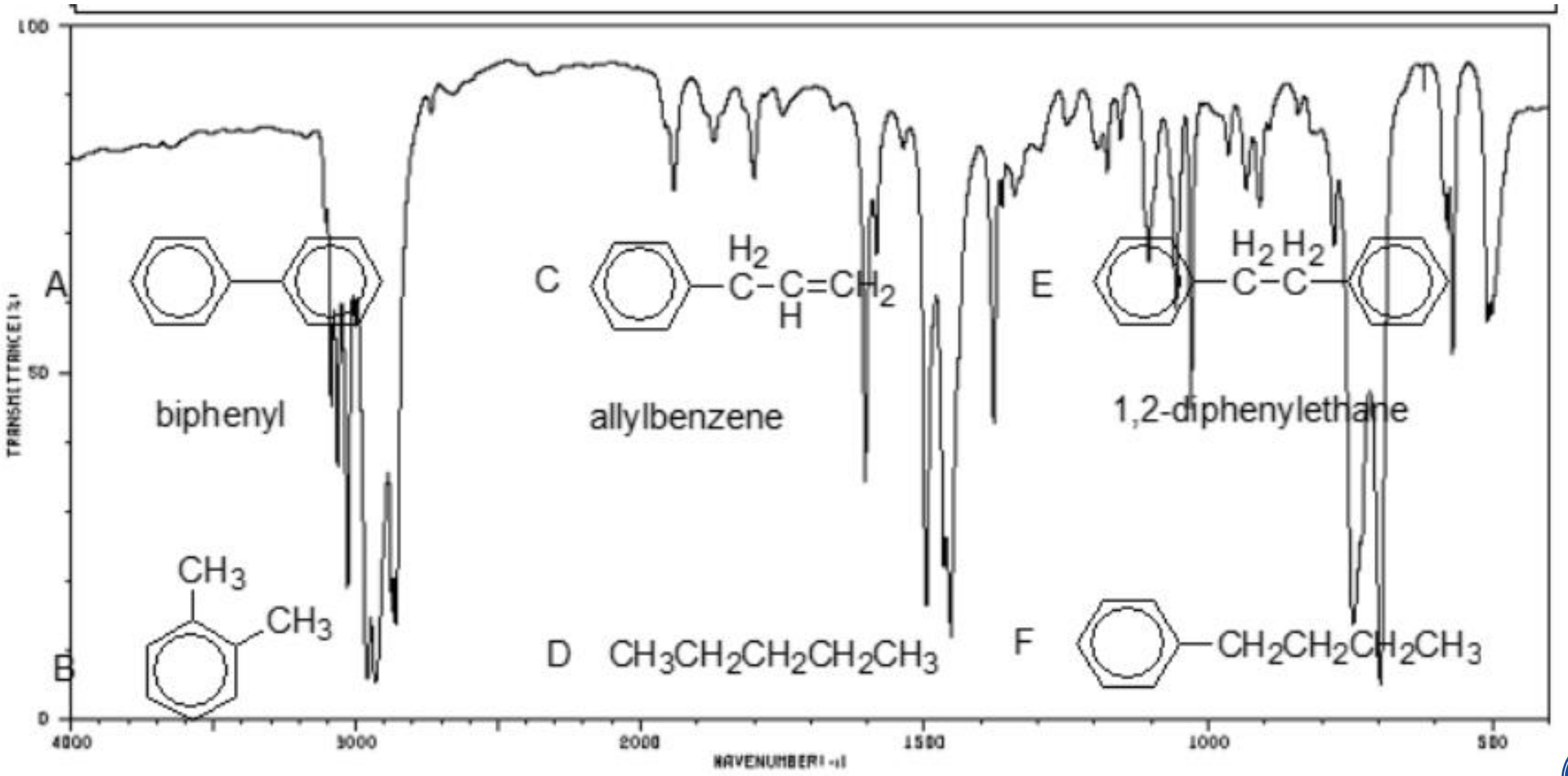
Matching Exercise



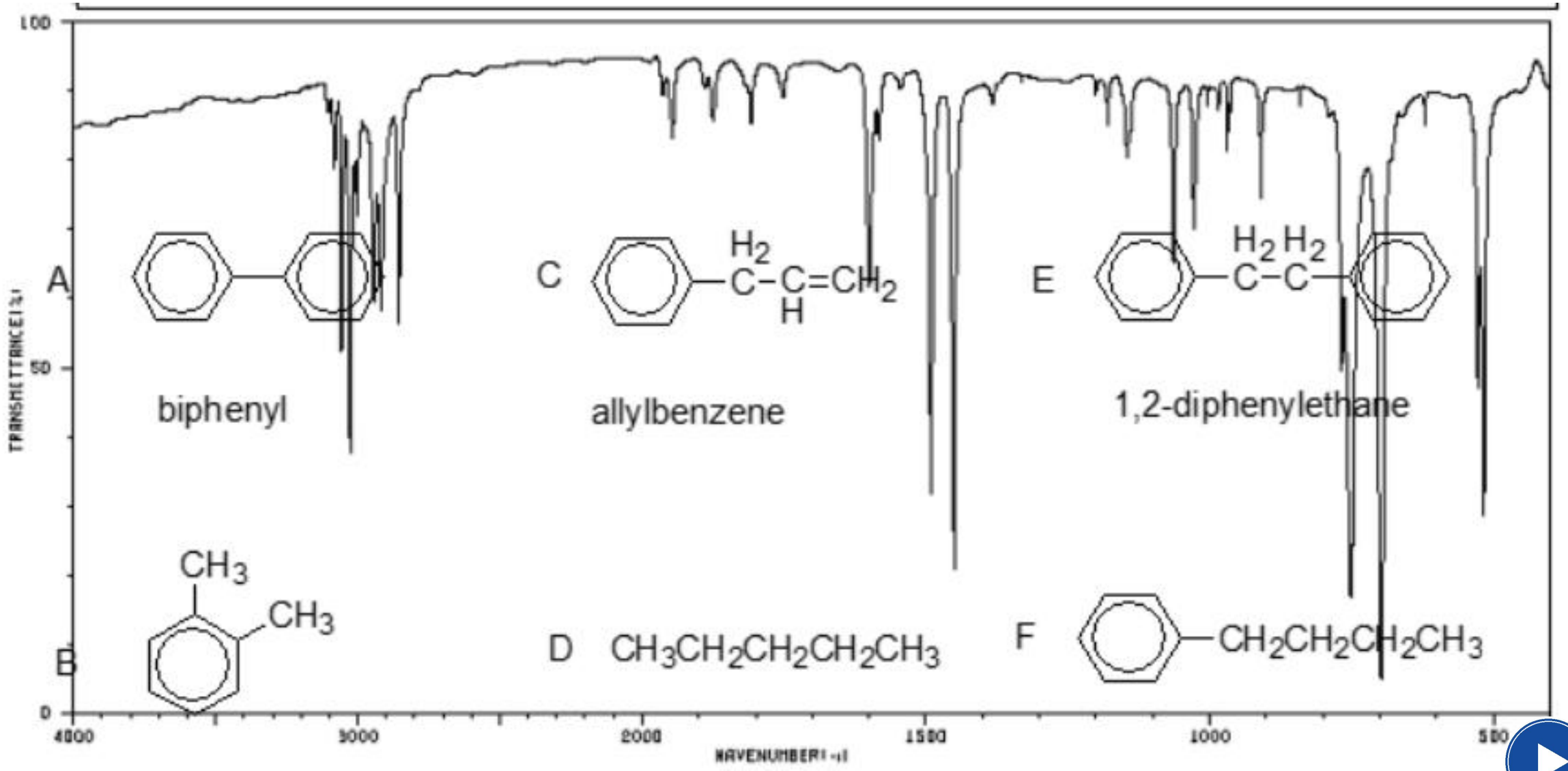
Matching Exercise



Matching Exercise



Matching Exercise



Master

- the basic principles of infrared spectrophotometry, the conditions for producing an infrared spectrum, the factors affecting peak position, and the characteristic and correlation peaks;
- the typical spectra of some organic compounds;
- given the structure of a compound, be able to write or sketch its infrared features.

Thank You

Next Chapter: Nuclear Magnetic Resonance Spectroscopy

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